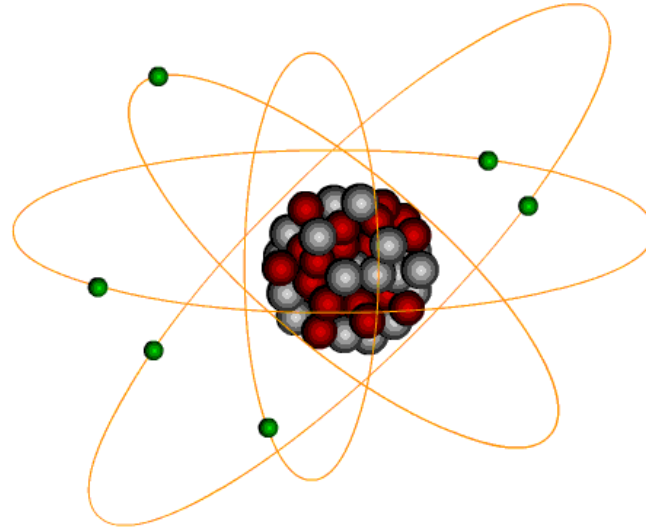




11.5 Inquires about Radioactivity to fulfill human needs.

Introduction to Radioactivity

- Radioactivity
 - Natural radioactive decay
 - Emission of α - particles
 - Emission of β - particles
 - Emission of γ -rays
 - Radioactive disintegration law
 - Graphical representation
 - Decay constant
 - Activity
 - Half life

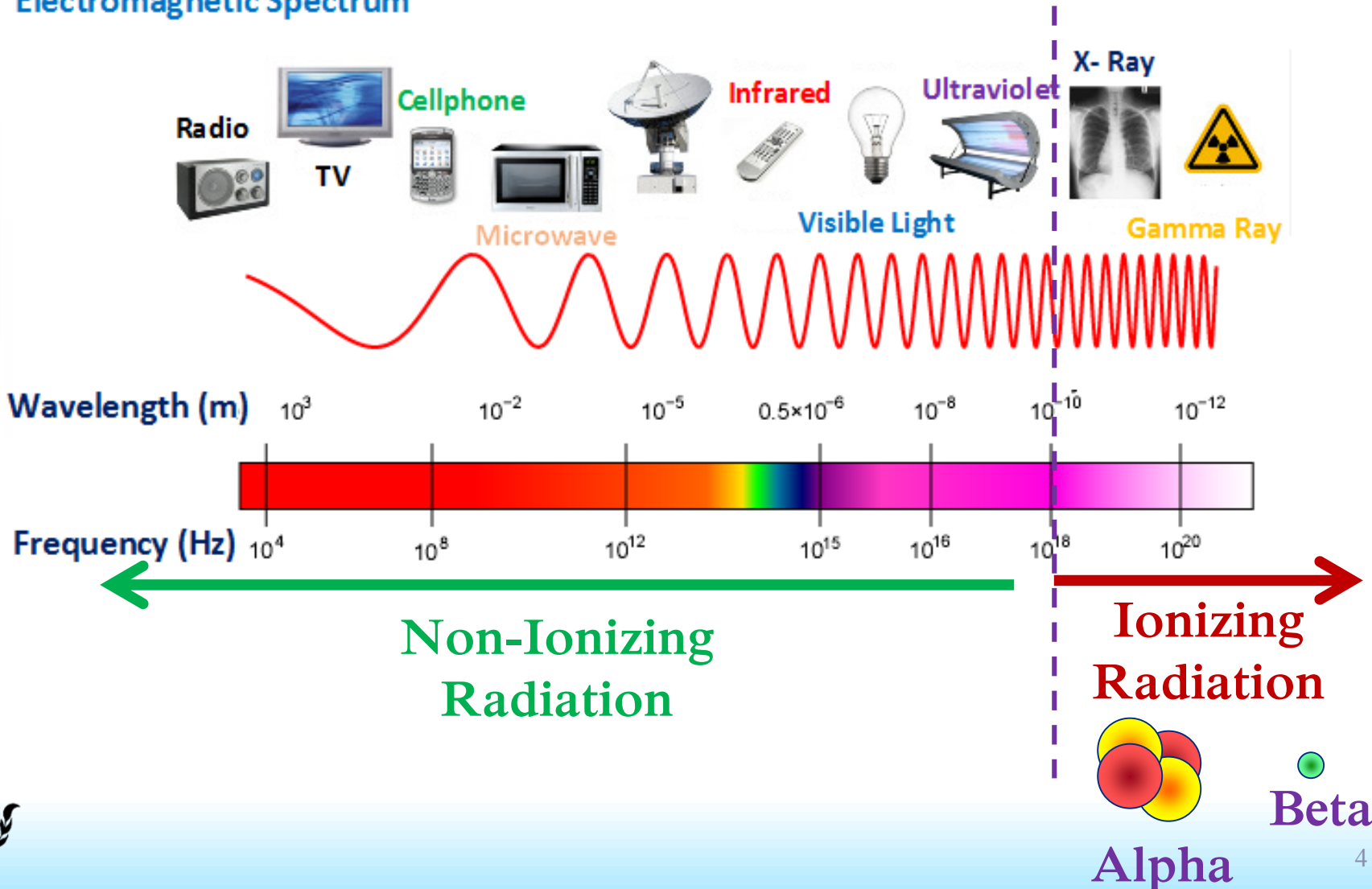
Radiation

- As a Ray or
- As a Wave or
- As a Particulate

Energy outflow

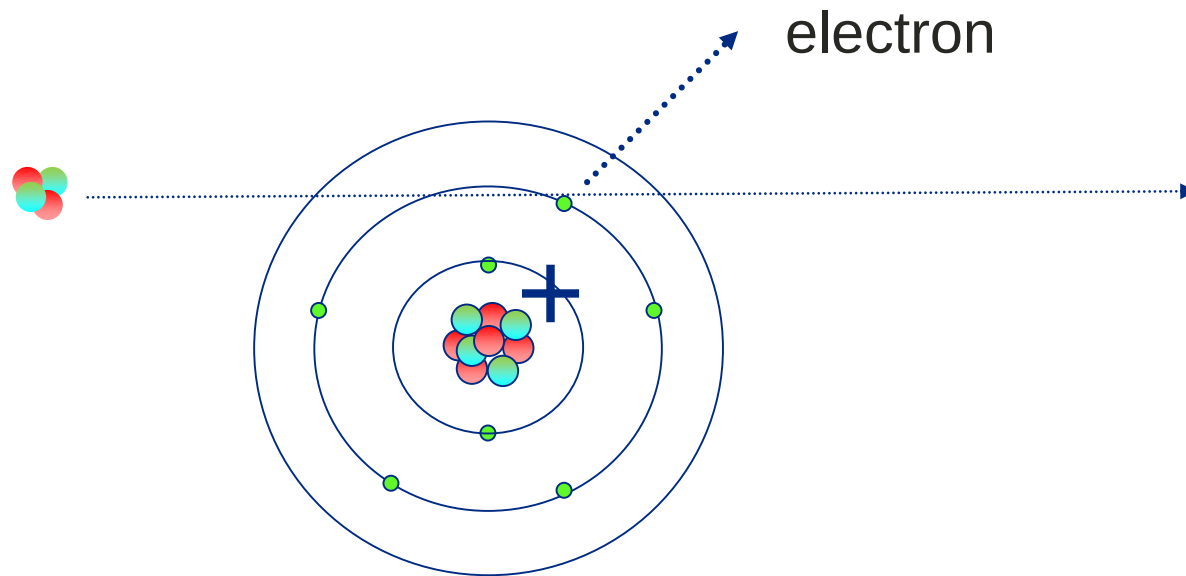
Ionizing Radiation

Electromagnetic Spectrum



Ionizing Radiation

Has enough energy to remove electrons from atoms –
ionization (34 keV)



Ionizing Radiation Types

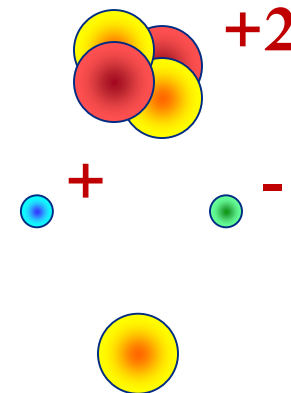
Rays (Electromagnetic Photons)

- X Ray Radiation
- Gamma Radiation (γ)



Partials

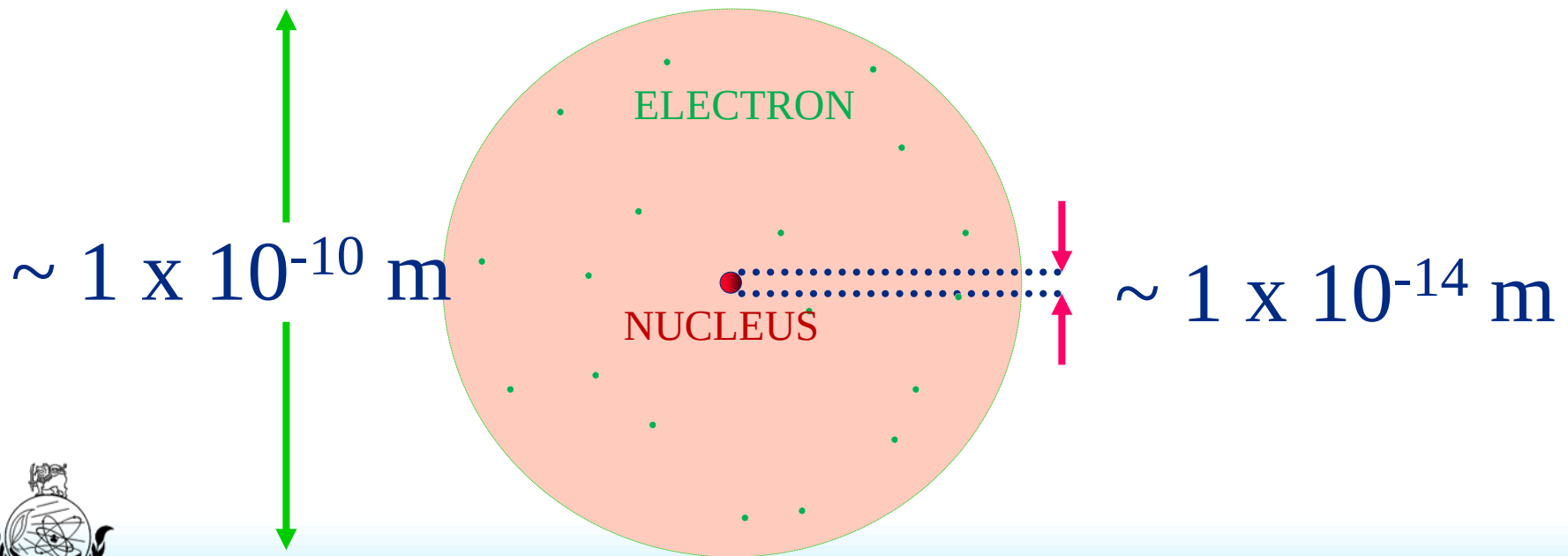
- Alpha (α)
- Beta ($\beta \pm$)
- Neutron Radiation (n)



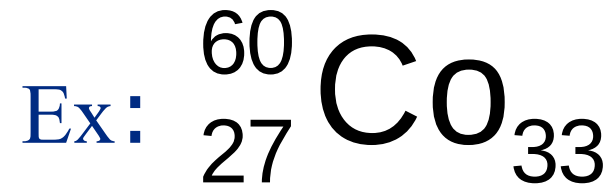
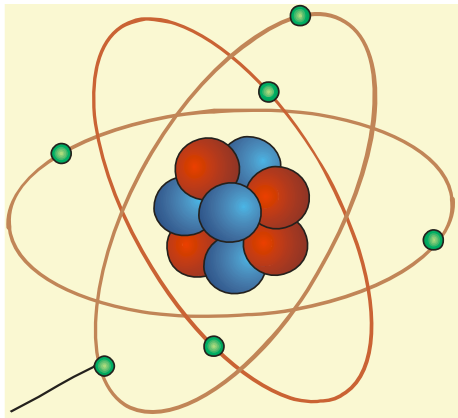
Atomic Structure

Mainly two parts

- NUCLEUS
- ELECTRONS



Symbolic presentation of the nucleus



Z – Number of Protons; Called as **Atomic Number**

Ex - Hydrogen $Z = 1$; Iron $Z = 26$; Uranium $Z = 92$

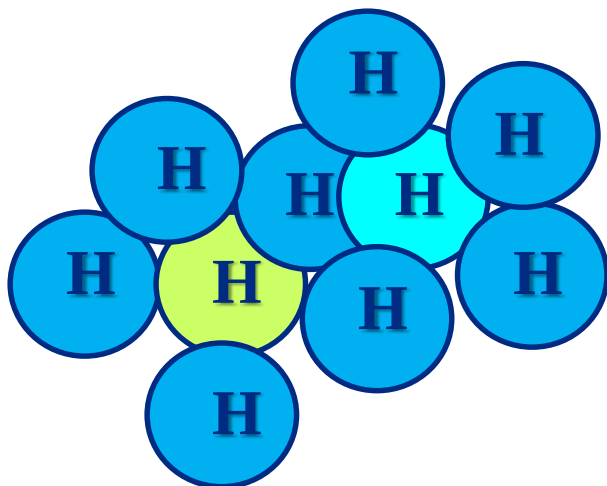
N – Number of Neutrons

A – Total number of Protons and Neutrons; Called as **Mass Number**

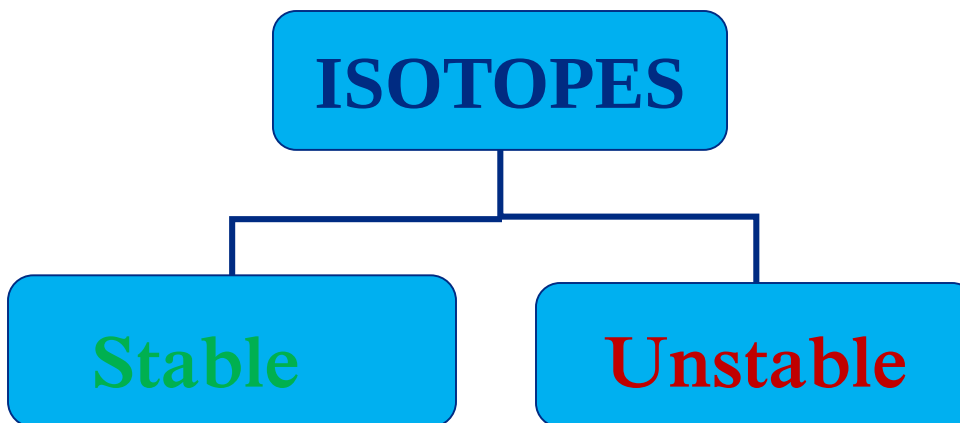
$$A = Z + N$$



Isotopes



- The different species of atoms in same element
- They have same number of protons (Z) but different number of neutrons (N)



Stable and Unstable Isotopes

- Unstable Isotopes Emit Ionization Radiation and try to Become Stable

^{58}Co

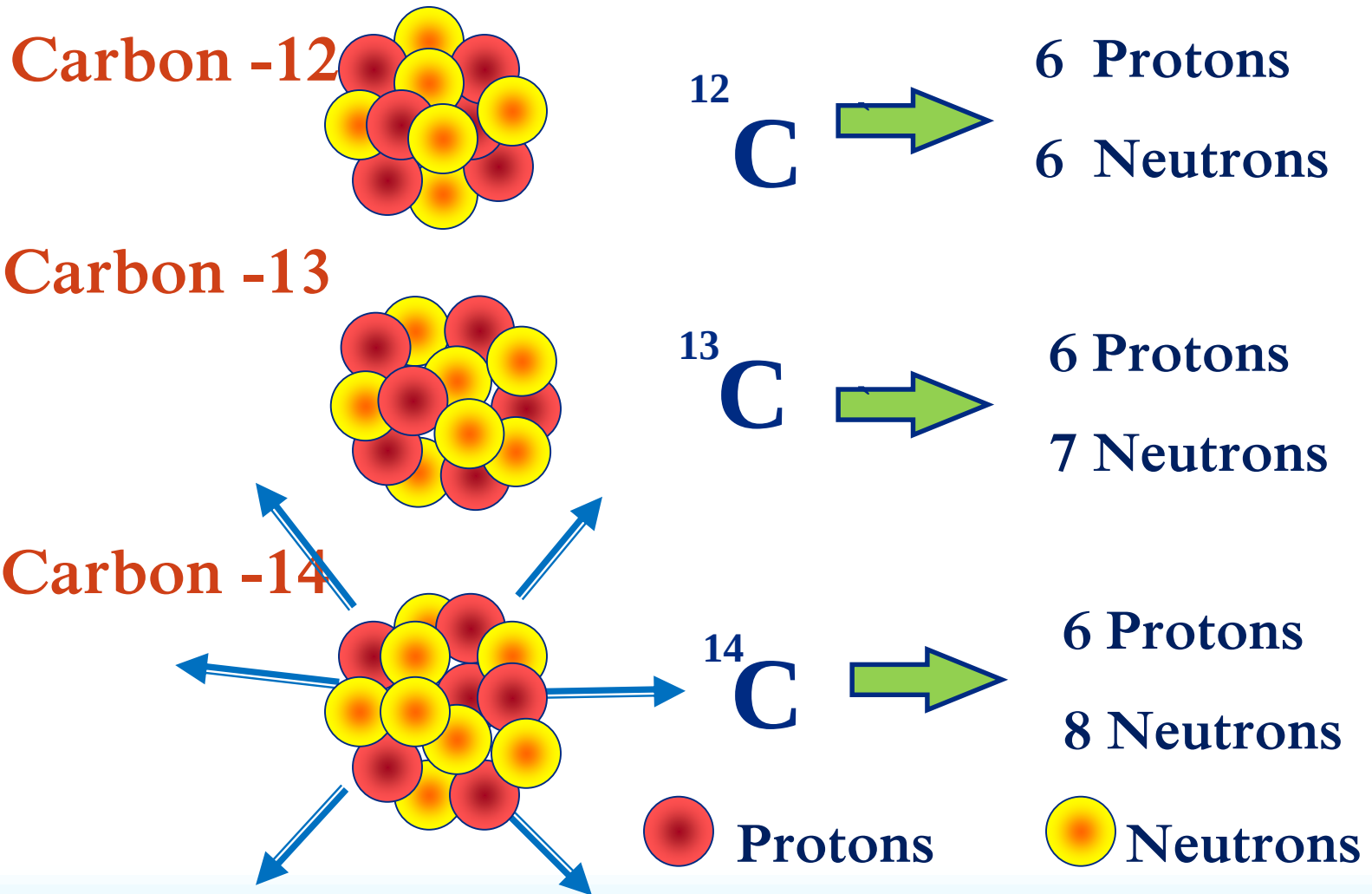
^{59}Co

Stable – Not Radioactive

^{60}Co

Unstable -
Radioactive

Isotopes



Radioactive Isotopes

Natural Radionuclides

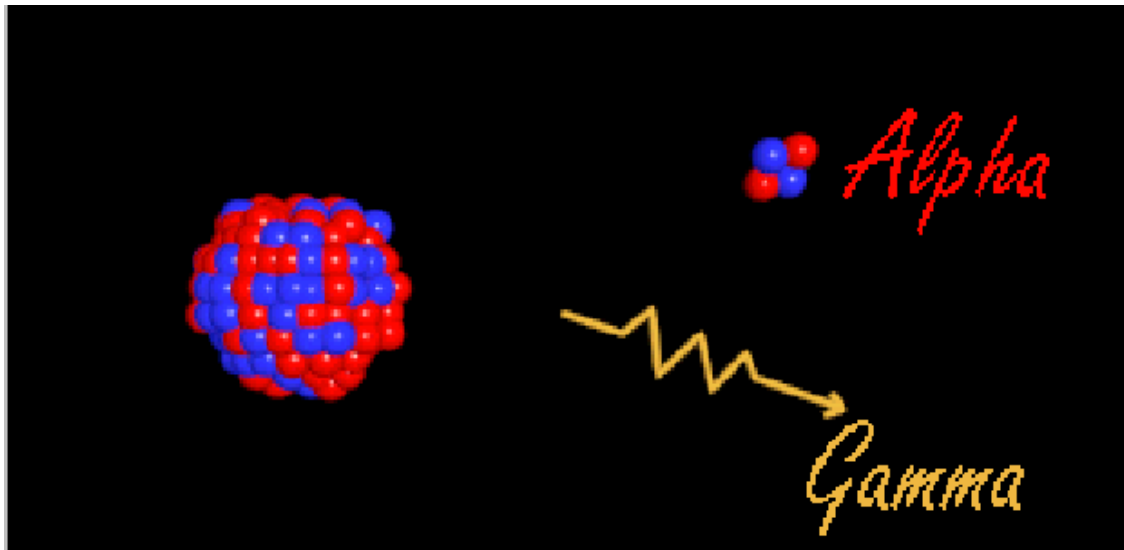
- K-40
- Th-232
- U-238
- Ra- 226
- Rn-222

Artificial Radionuclides

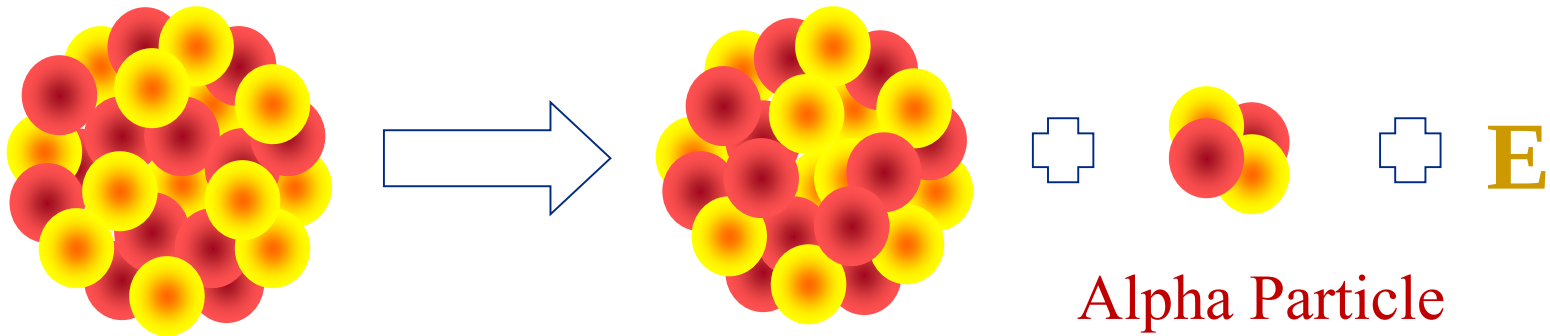
- Cs-137
- Co-60
- I-131
- Sr-90
- Am-241

(α) Alpha Radiation

- For nuclei with $Z > 82$ alpha decay is also a possibility.
- Alpha particles are heavy charged particles consisting of two neutrons and two protons, or the nucleus of the helium-4 atom

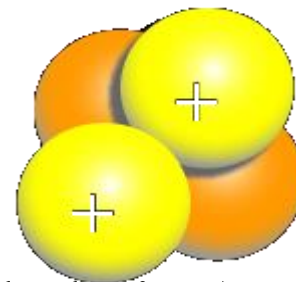


Alpha Emission

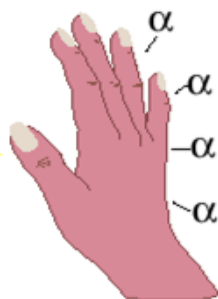


Properties of Alpha(α) particles

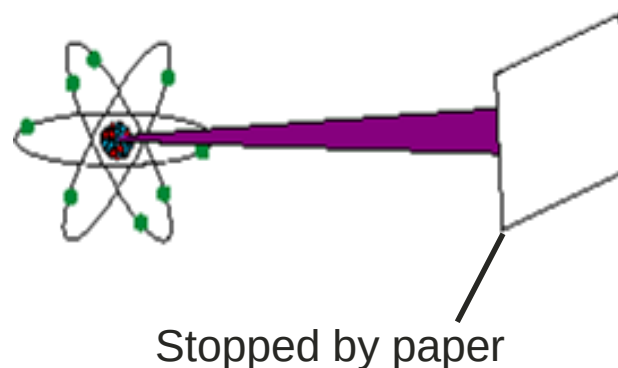
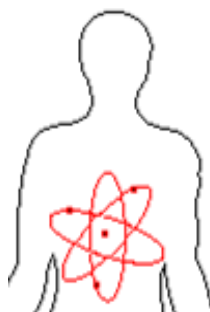
- Consists of 2 Protons and 2 Neutrons
- Relatively big and heavy
- Can be stopped by a sheet of paper or skin
- Danger if incorporated into the body (ingestion, inhalation)
- It can travel only 2-3 cm in Air



Cannot get
through
skin

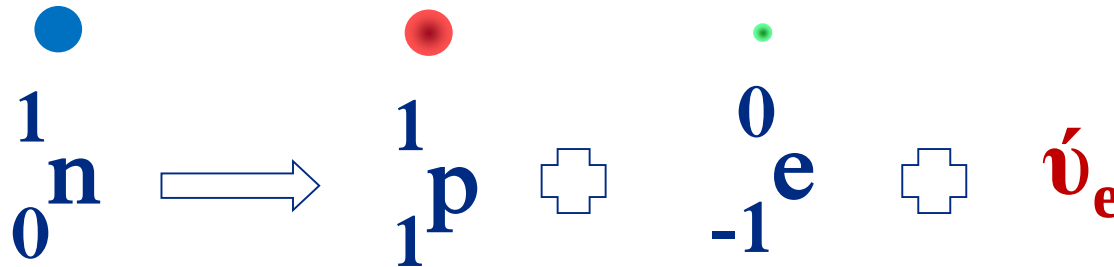
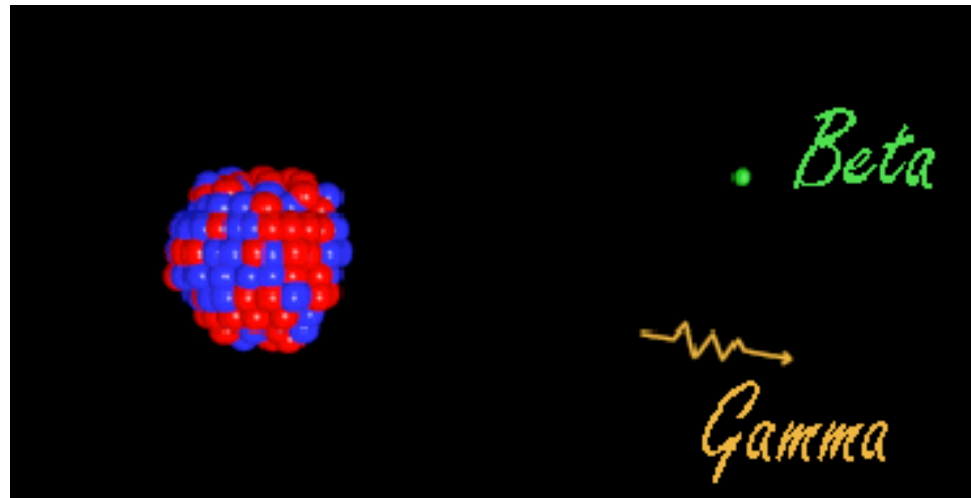


Internal
hazard

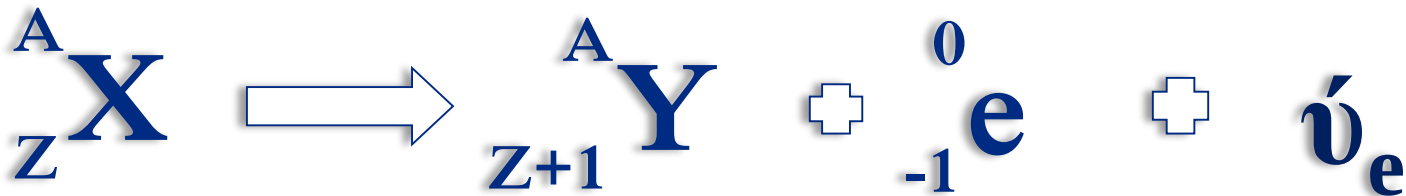
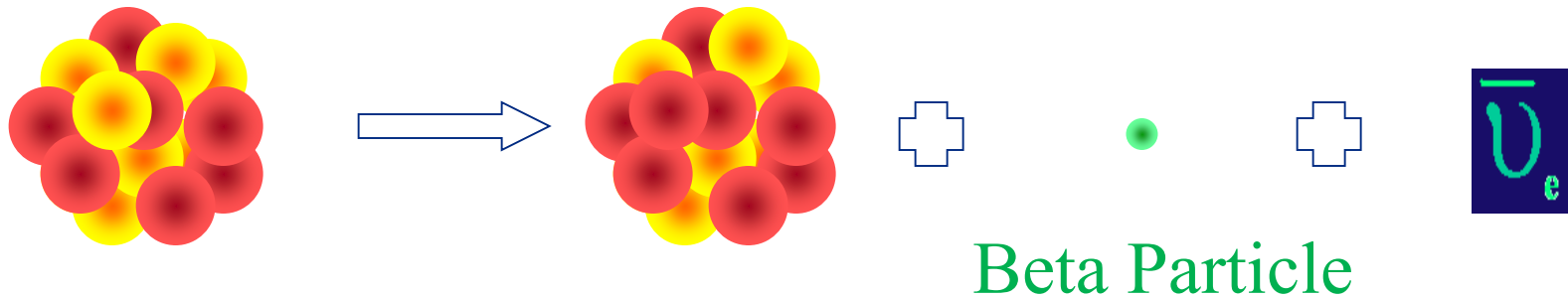


$(-\beta)$ Beta Radiation

If nucleus has too many neutrons, then it will undergo radioactive decay by emitting a $-\beta$ particle (an electron) and an anti-neutrino,

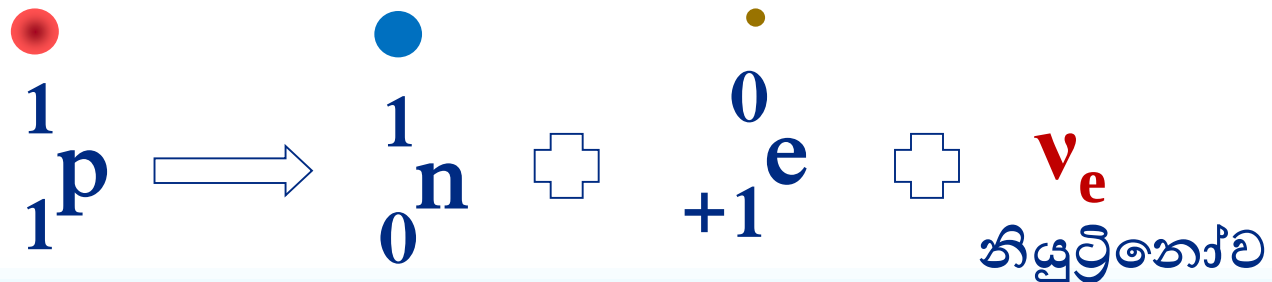
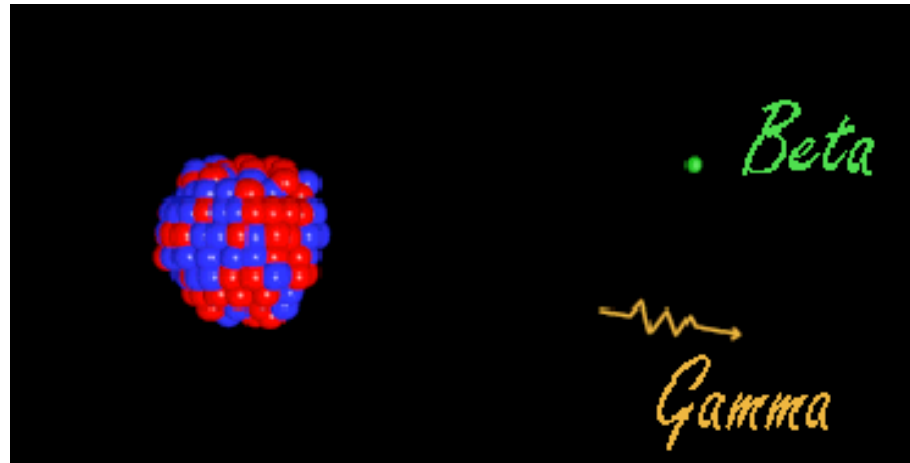


Beta ($-\beta$) Emission

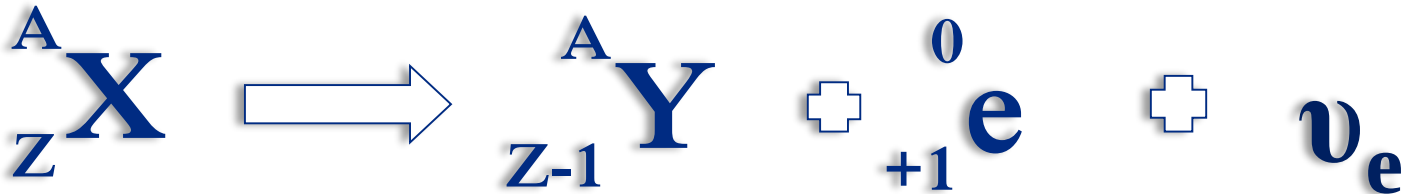
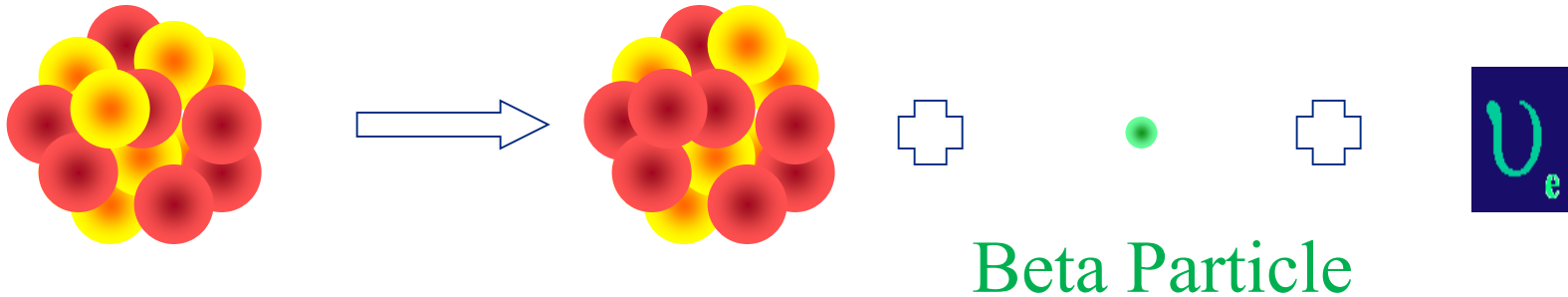


(+β) Beta Radiation

If the nucleus has too many protons, then it can either capture one of the atomic electrons (usually from the inner most or k - shell electron) or if there is enough energy available, it can emit a positron or β^+ particle.

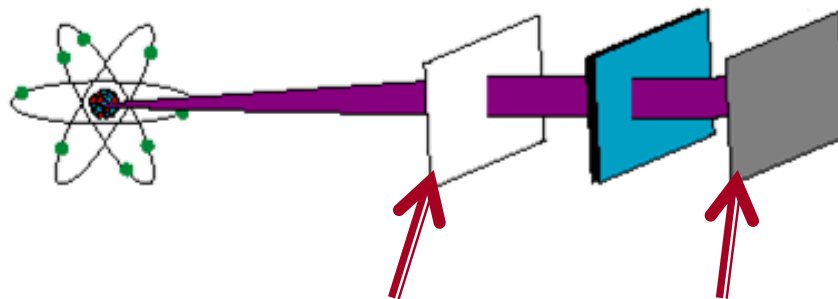


Beta (+ β) Emission



Properties of Beta Particles

- Smaller than the Alpha particle and charge also less, Therefore ionization power is less than Alpha
- Beta penetrate through paper, cardboard, wooden, but can stop by thick Aluminum
- It travels 10- 15 cm in Air
- Beta can effect to skin or organ near the skin.

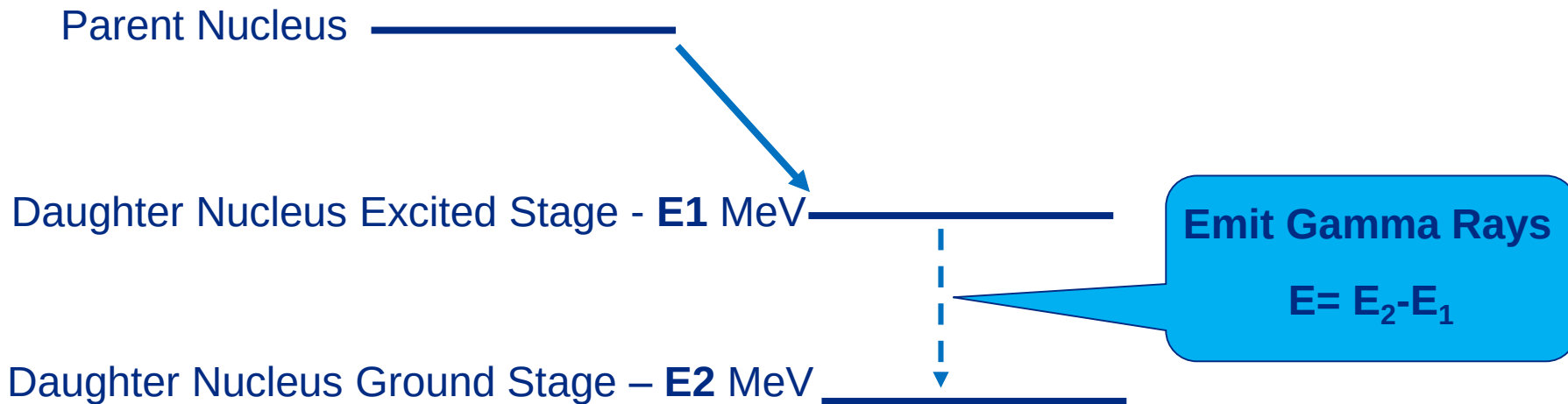


Paper

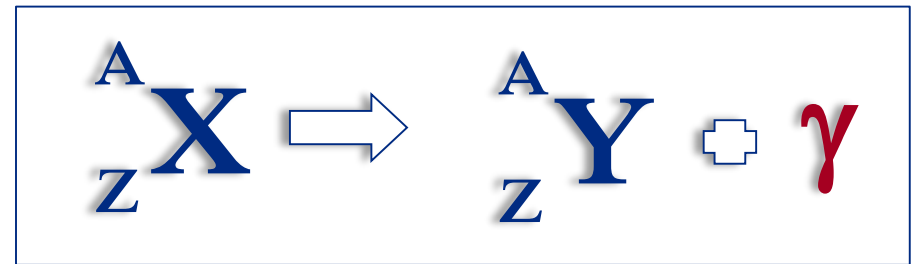
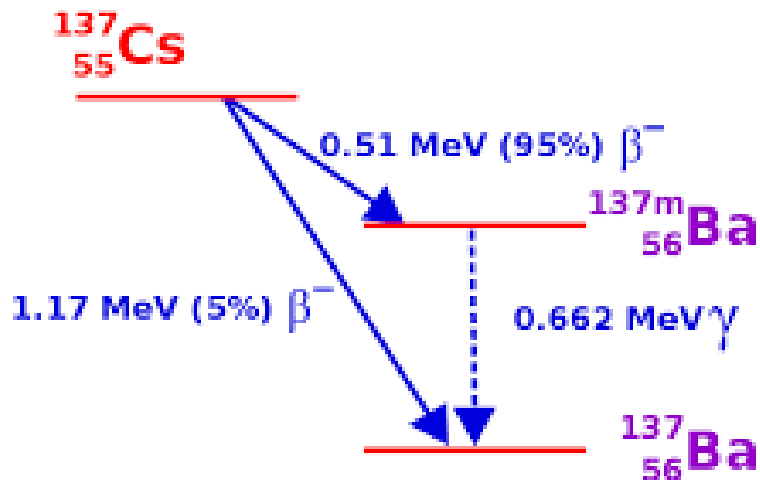
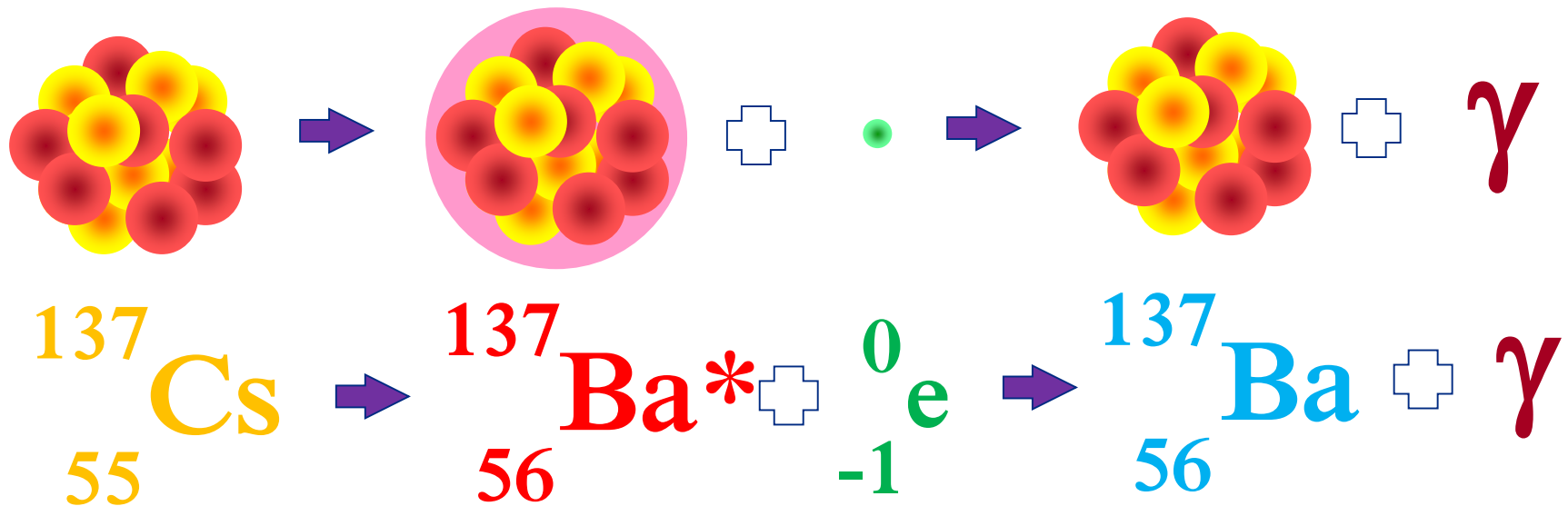
Aluminum

Gamma Radiation (γ)

- Gamma is High energy Electro Magnetic wave
- Most radioactive decay is not to the ground state of the daughter nucleus.
- Instead the daughter nucleus is usually left in an an excited state and then de-excites or moves to a lower energy level by emitting photons (gamma rays) until it reaches the ground state.

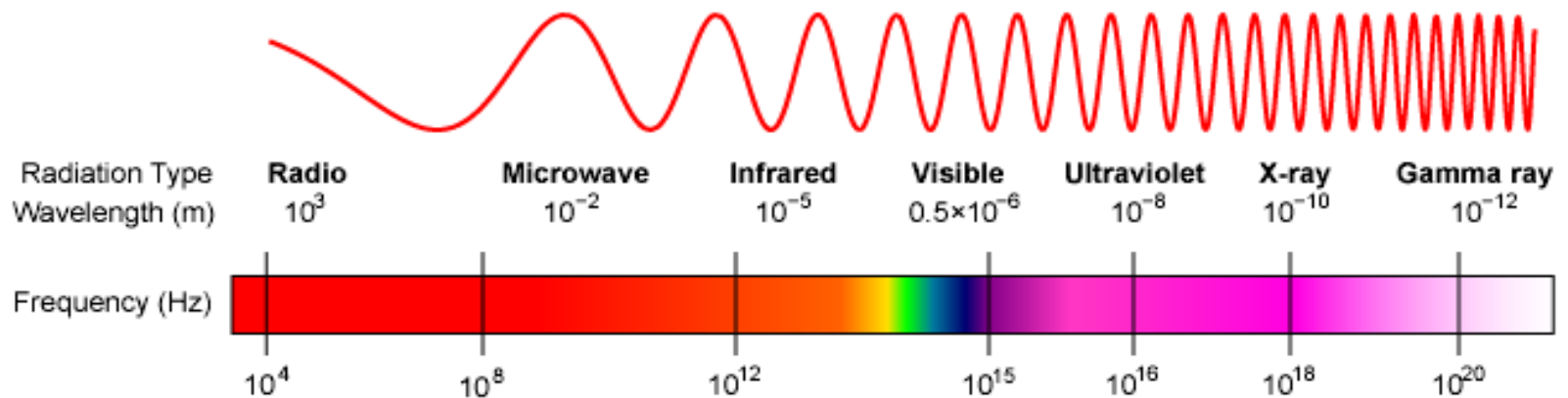


Gamma Radiation Emission



Properties of Gamma Radiation

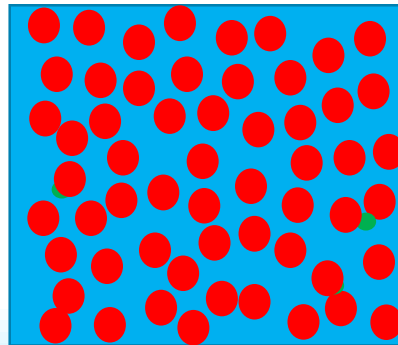
- As it has no change, ionization power is low but penetration power is high.
- Can stop with thick Lead (Pb) or Concrete.
- External Gamma Source can damage to internal organs in the body
- Can be detected easily.





Radioactive Decay

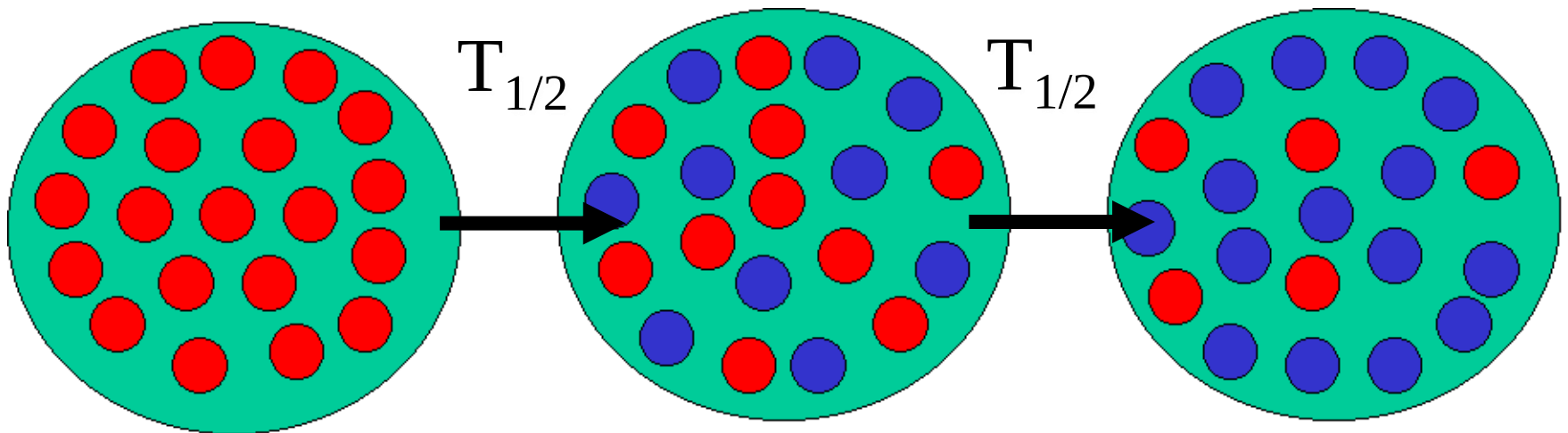
- Unstable nuclei release energy to become more stable nuclei is called as a Radioactive Decay
- Released Energy emit as Radiation, it may be a particle or electromagnetic wave
- Decay of one nuclei is called as a disintegration



Half life ($t_{1/2}$)

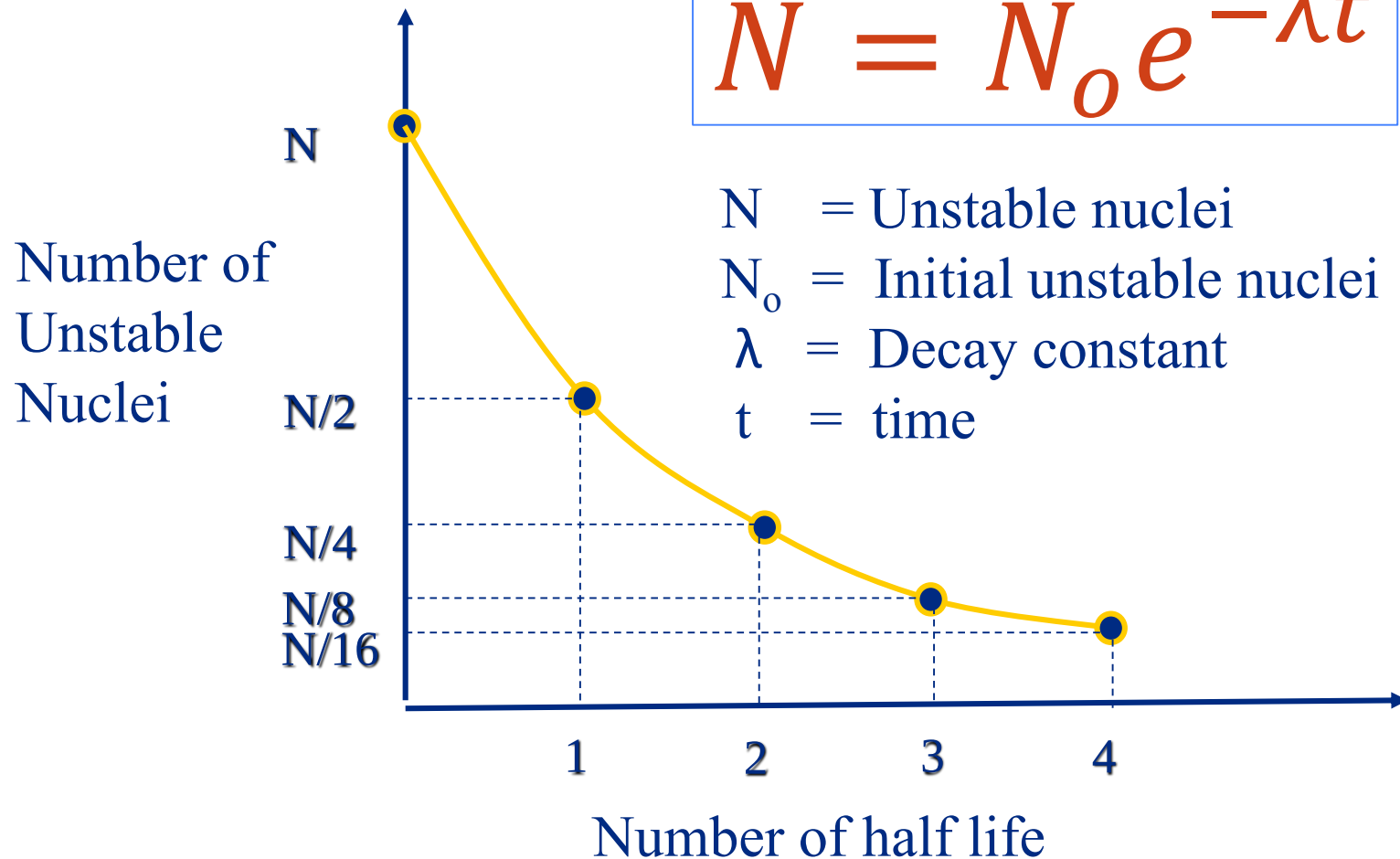
The half life is defined as the period of time that it takes for one half of the radioactive nuclei to decay

$t_{1/2}$ is the half life of the radioactive material.



Graphical Representation

$$N = N_0 e^{-\lambda t}$$



Decay Constant (λ)

$$N = N_0 e^{-\lambda t}$$

$$\lambda = \frac{0.693}{t_{1/2}}$$

$$\frac{N_0}{2} = N_0 e^{-\lambda t_{1/2}}$$

$$\frac{1}{2} = e^{-\lambda t_{1/2}}$$

$$\lambda t_{1/2} = \ln [2]$$

$$N = N_0 e^{-\frac{0.693}{t_{1/2}} t}$$

N = Unstable Nuclei

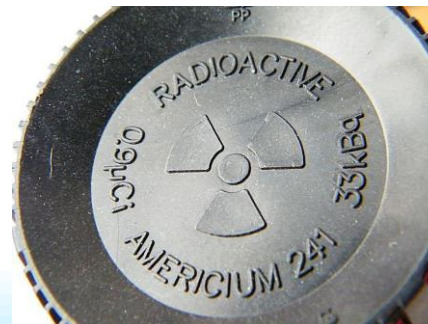
N₀ = Initial unstable nuclei

t = Time

t_{1/2} = Half life

Activity (A)

- The activity, A, or number of nuclei decaying (disintegrating) per unit time is given by
$$A = \lambda N$$
- where N is the number of radioactive nuclei and λ is the decay constant, or probability per unit time that a nucleus will decay.
- The units of activity are **Becquerel (Bq)** or **Curie (Ci)**.
- One Bq is defined as one disintegration per second while a Ci is defined as 3.7×10^{10} disintegrations per second



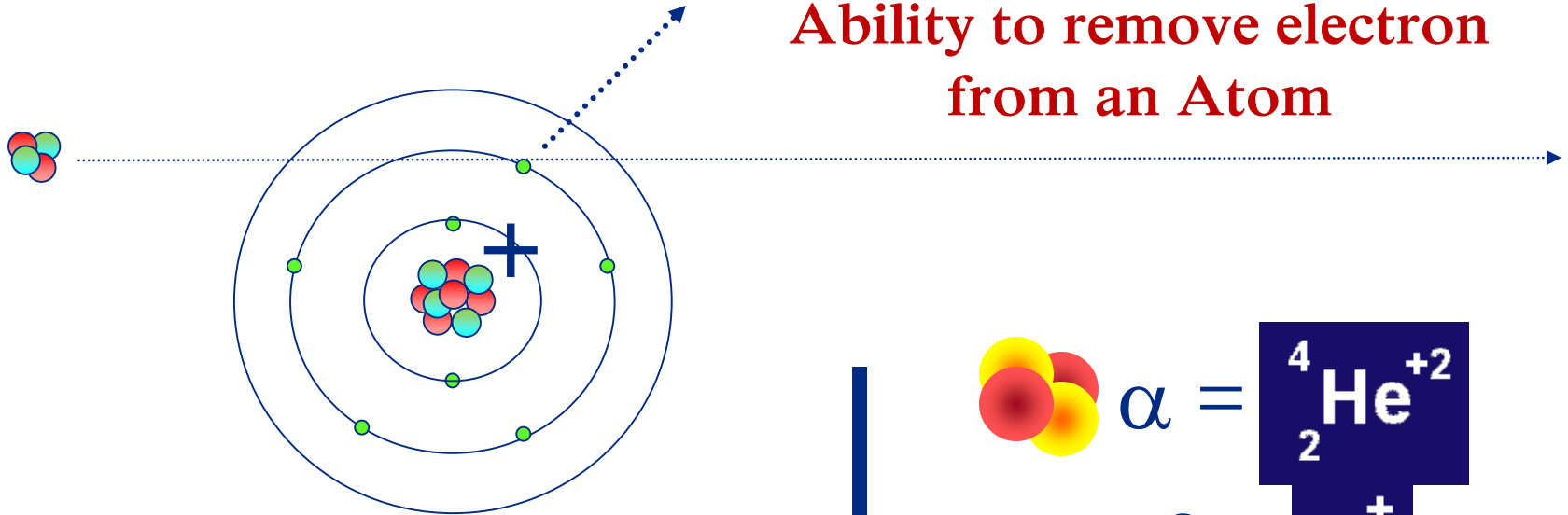
Half Life ($t_{1/2}$)

Natural Isotopes	Half-life ($t_{1/2}$)
^{235}U	$7.04 \times 10^8 \text{ yr}$
^{238}U	$4.47 \times 10^9 \text{ yr}$
^{232}Th	$1.41 \times 10^{10} \text{ yr}$
^{226}Ra	$1.60 \times 10^3 \text{ yr}$
^{222}Rn	3.82 days
^{40}K	$1.28 \times 10^9 \text{ yr}$

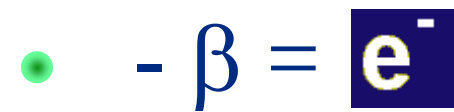
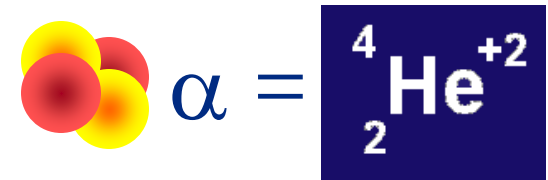
Artificial Isotopes	Half-life ($t_{1/2}$)
^3H	12.3 yr
^{131}I	8.04 days
^{129}I	$1.57 \times 10^7 \text{ yr}$
^{137}Cs	30.17 yr
^{90}Sr	28.78 yr
^{99}Tc	$2.11 \times 10^5 \text{ yr}$

Ionization Power

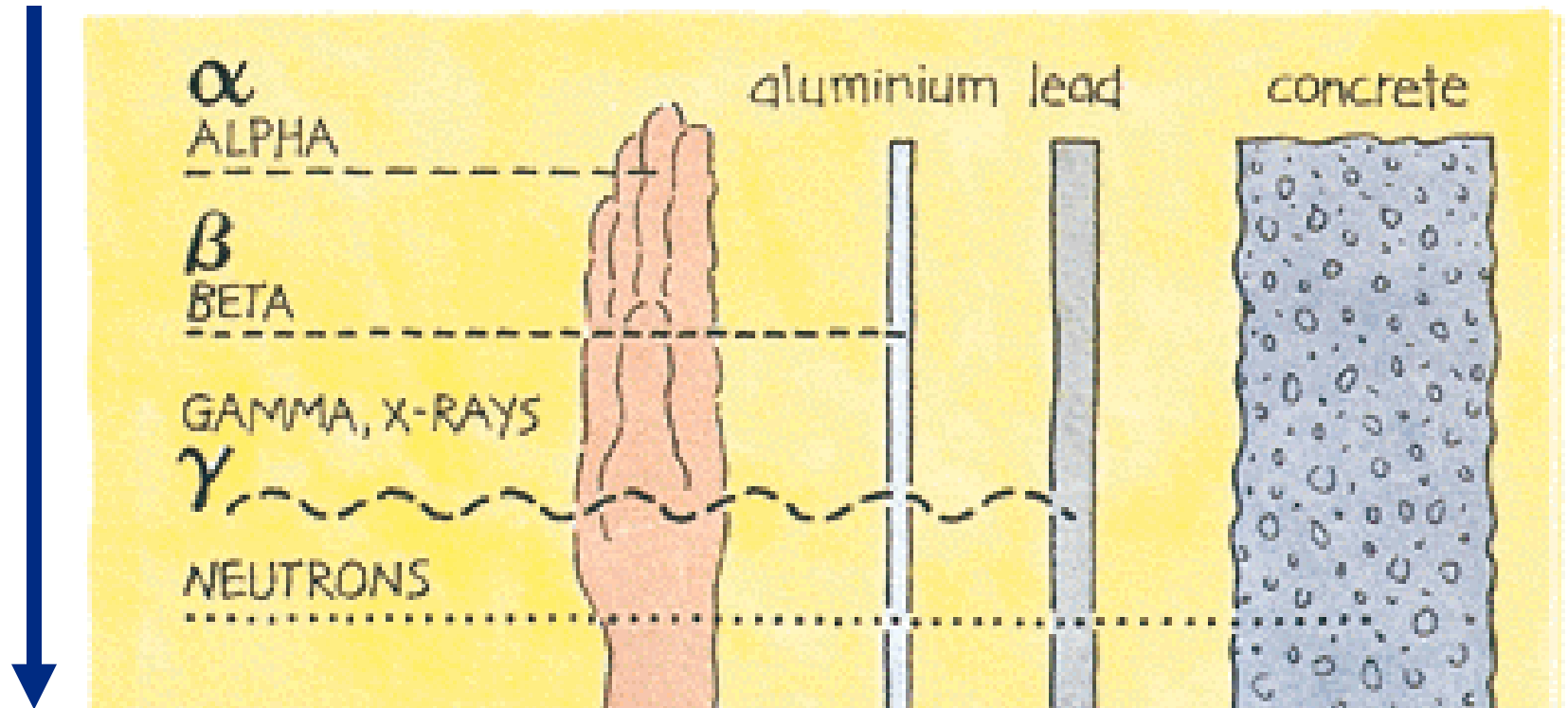
Ability to remove electron from an Atom



Reduce the Ionization Power



Penetration Power

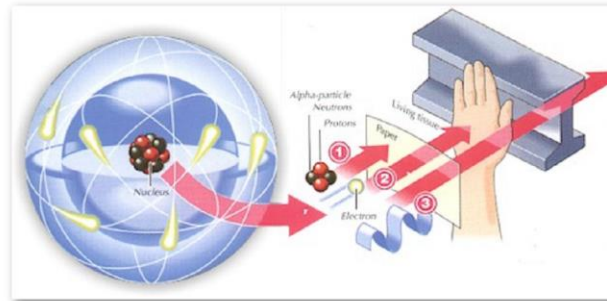


Increase the penetration power

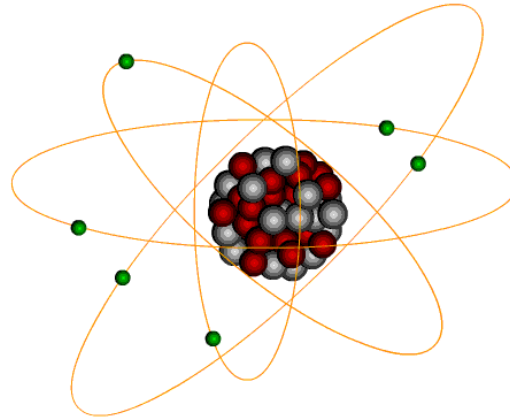
Nuclear Technology

Based on the properties of Ionization Radiation

- Ionization ability,
- Penetration ability
- Energy difference of radiation emitted from different isotopes



Ionization radiation is harmful to the human body, so it should be used with controlled manner

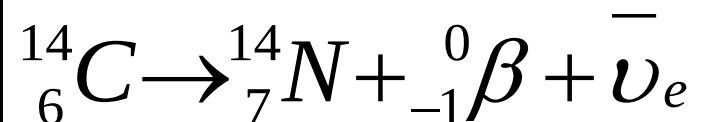
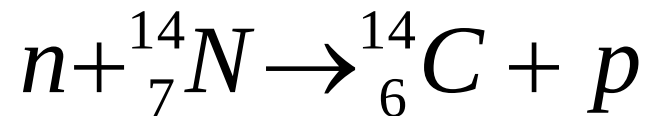
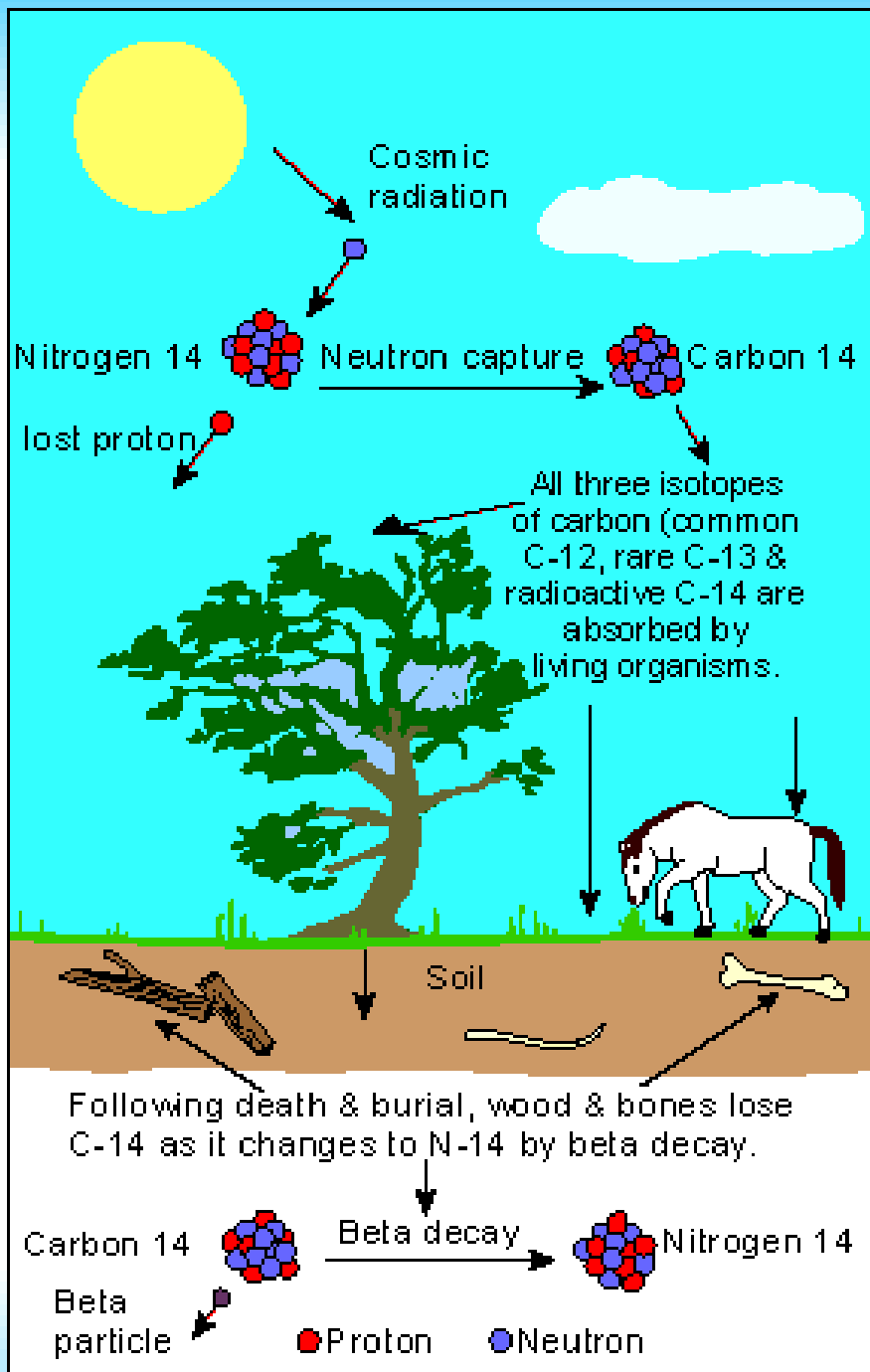


Use of Radioactivity

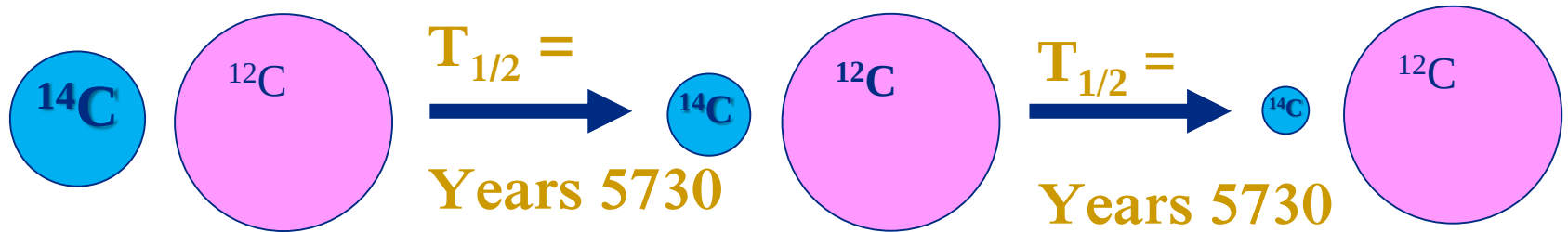
- Uses of radioactivity
 - Radioactive dating (C^{14})
 - Uses of radioactivity in
 - Medicine
 - Engineering
 - Agriculture
 - Environment & Reserch

Radioactive dating (^{14}C - Carbon Dating)

- ^{14}C Carbon dating is used for prediction of died date of living objects
- Cosmic Radiation comes from sun & galaxies, reacts with Nitrogen -14 (^{14}N) in the upper atmosphere and produce carbon -14.
- ^{14}C is unstable nuclei and it decays. but due to continues production of ^{14}C , ratio of $^{14}\text{C} : ^{12}\text{C}$ in the Atmosphere is constant
- Living organisms absorb Carbon from Atmosphere. Therefore $^{14}\text{C} : ^{12}\text{C}$ Ratio of living organisms are also constant.
- But when the organism died, no carbon exchange with the atmosphere, As the ^{14}C decay the ratio of $^{14}\text{C} : ^{12}\text{C}$ change with the time.



- This technique can be used to calculate the age of (fossils) dead parts of trees and animals
- This method can be used only for predict the age of fossils more than 2500 years old. Less agers uncertainty is too high as the Carbon -14 half life is 5730 years



$$N = N_0 e^{-\lambda t}$$

N = Number of Carbon 14 (^{14}C) atoms remain after t time

N_0 = Initial number of Carbon -14 (^{14}C) atoms

t = Time

λ = Decay constant = $0.693 / t_{1/2}$

$t_{1/2}$ = Half life

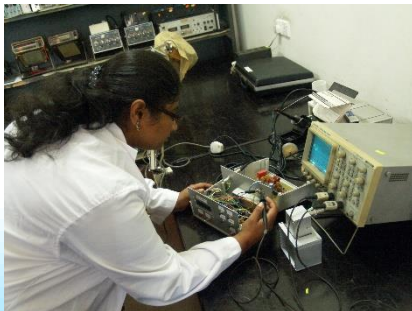
Applications of Nuclear Technology in our Country

Currently nuclear technology used in ,

- **Medical**
- **Agricultural**
- **Industrial**
- **Environmental protection and Research fields in Sri Lanka**



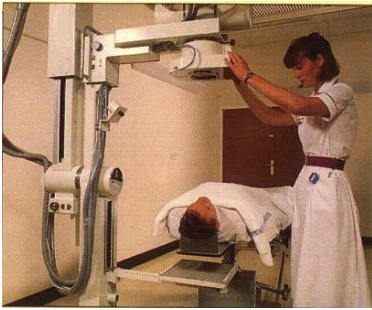
Generation of nuclear power is one of our future expectation.



Medical Applications

Radiation used for Diagnostic purposes

- X-Ray imaging
- Fluoroscopy and Mammography investigations
- CT-Scan with X-ray
- Gamma Camera Imaging
- Prediction of diseases like Denga, Filariasis, Rat-bite fever with Radioimmunoassay



Medical Applications

Radiation is used for Treatments

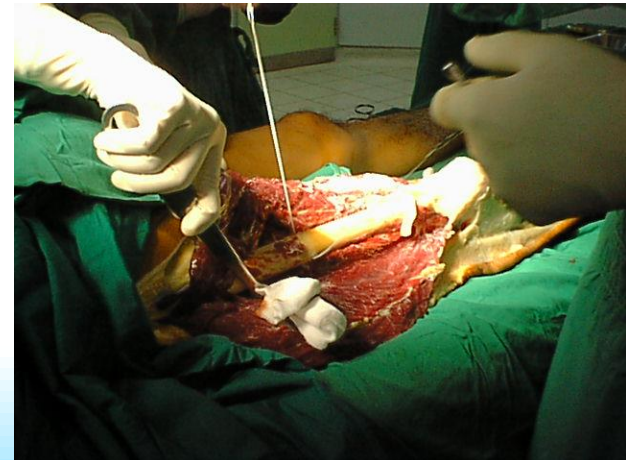
Cancer treatments

Tele therapy

Brachytherapy



Sterilization of human tissues before the transplant



Agriculture Application

- Absorption efficiency of fertilizer
- Physiological studies in plants
- Improving the production and quality of farm animals
- Determination of chemical residues in animal and aquatic products
- Pest control with sterile insect technique
- Improved breeding varieties
- Obtaining soil erosion data
- Soil moisture measurement
- Minimization of food preservation and post-harvest damage
- For plant quarantine



Industrial Applications

None Destructive Testing (NDT)

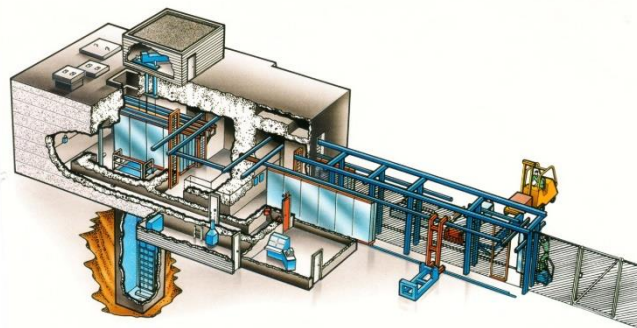
- These techniques are used to identify the manufacturing defects and operation defects (Crack, Slag) of the metal parts.
- NDT support to provide quality services by avoiding accidents at aircraft, ship, bridge, turbine, boilers, etc.



Industrial Applications

Gamma Irradiation is used for

- Sterilization of Medical Products
- Preservation of food
- Increase the value of Gems (by Gem Irradiation)
- Production of quality goods by polymerization

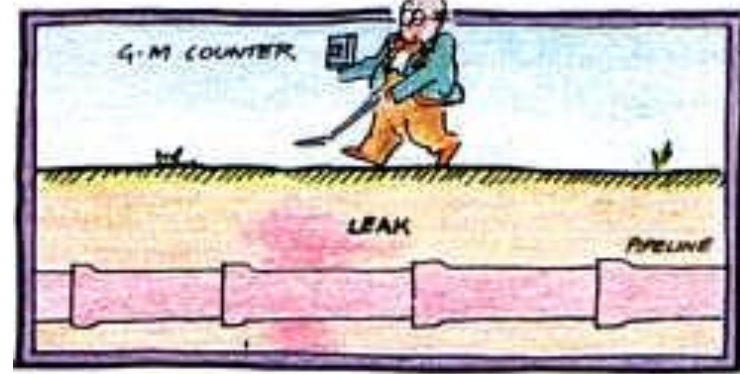


COBALT-60 IRRADIATION FACILITY
WITH CONVEYOR SYSTEM (CANADIAN IRRADIATION CENTER)



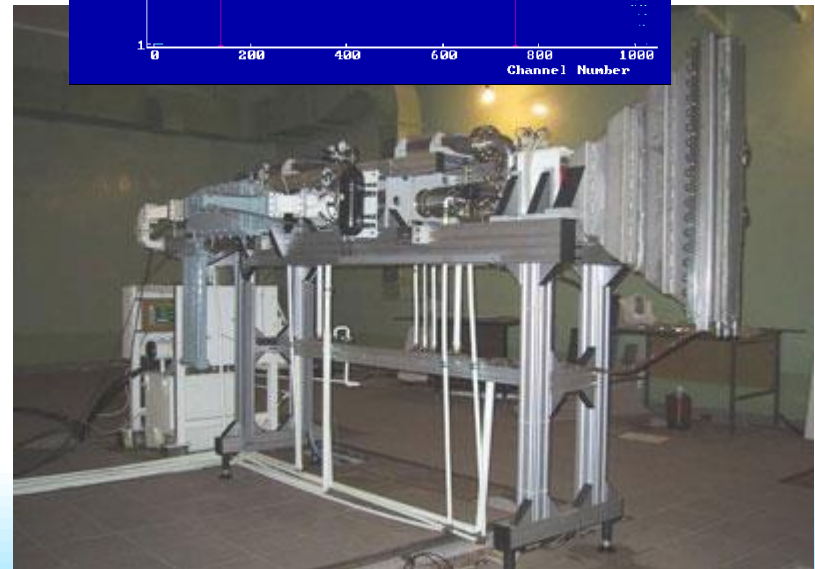
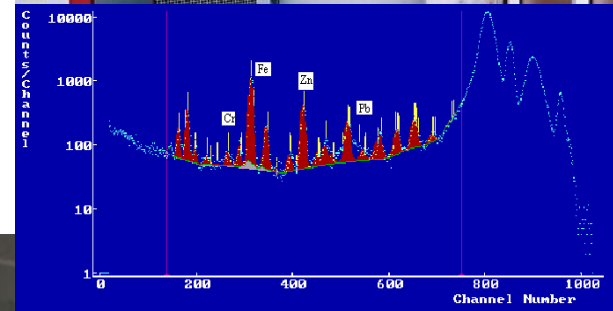
Industrial Applications

- Identification of leakages and obstacles in underground petroleum pipes, waste tubes etc.
- Nuclear gauges
automated industrial systems helps for quality control and safe operation.



Environmental Protection

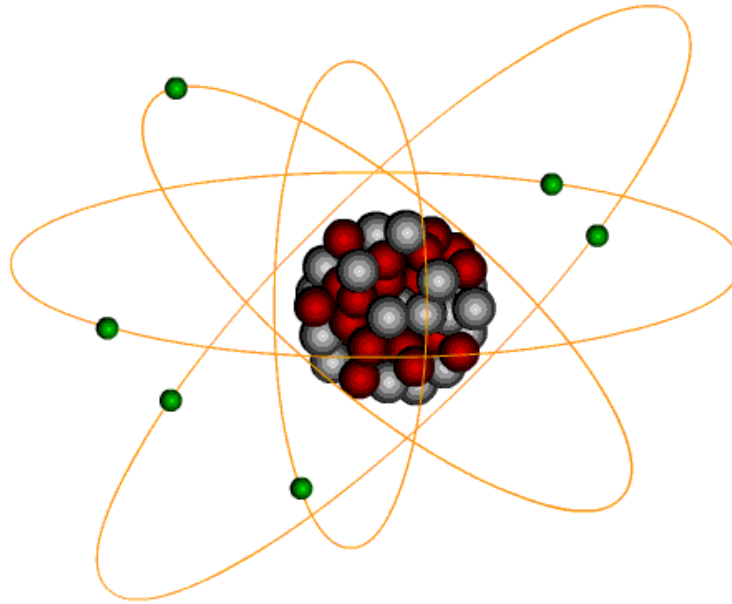
- Nuclear Analytical Techniques
XRF and TXRF analytical techniques used to identify heavy metals in Air, Water, Soil samples to control environmental pollution
- Electron beams
 - Purification of waste water emit from industry
 - Precipitation of dust particles emit from coal power plants



Research Applications

- Nuclear Analytical Techniques –
ICPMS (Inductively coupled plasma mass spectrometry) , IRMS (Isotope-ratio mass spectrometry) used for identify the isotopes in a sample
- Oil Exploration
- Isotope Hydrology –
Groundwater study with variation of isotopic ratios
Gather data for safety of reservoirs and dams

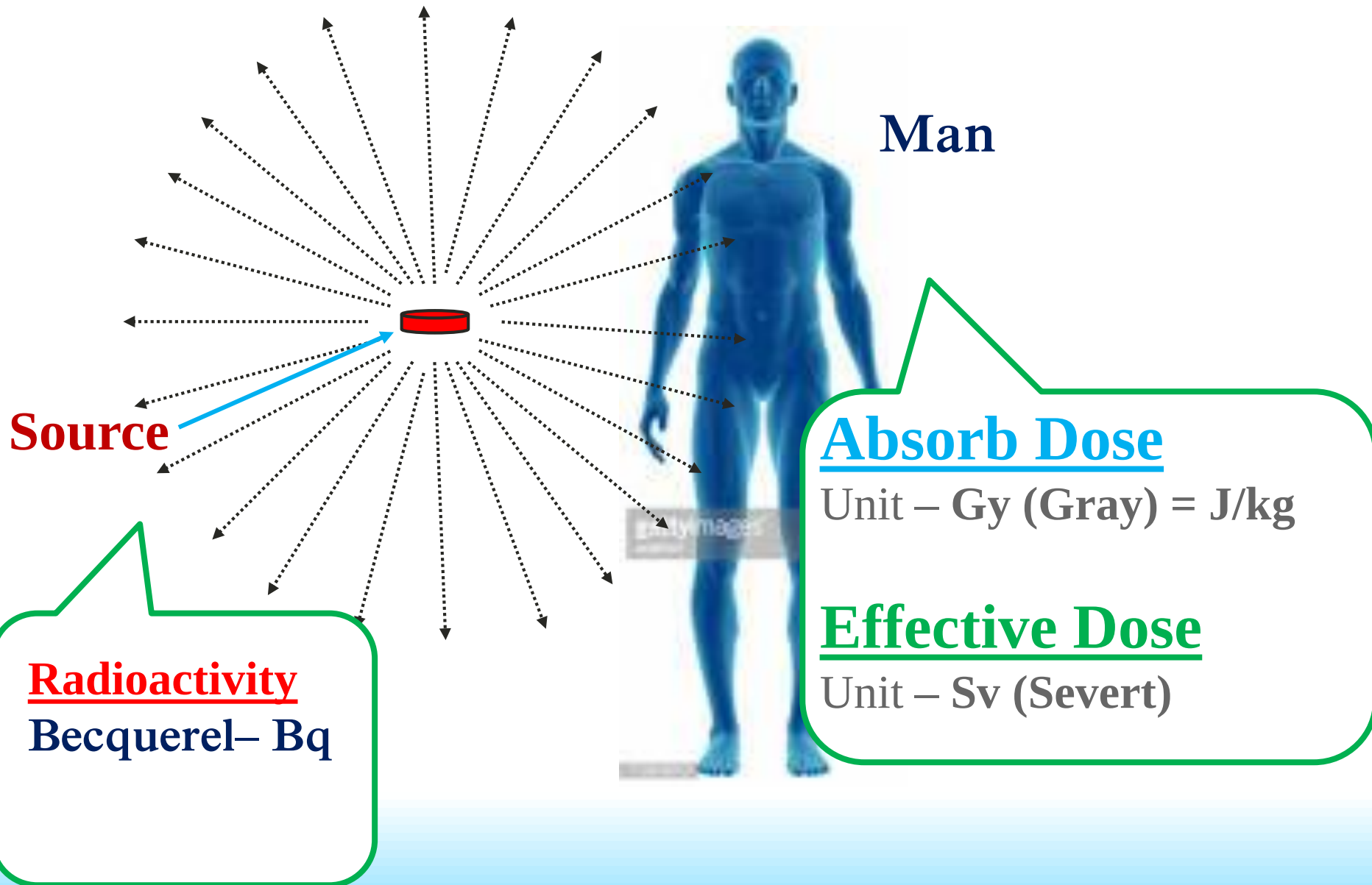




Health hazards of radiation and safety precautions

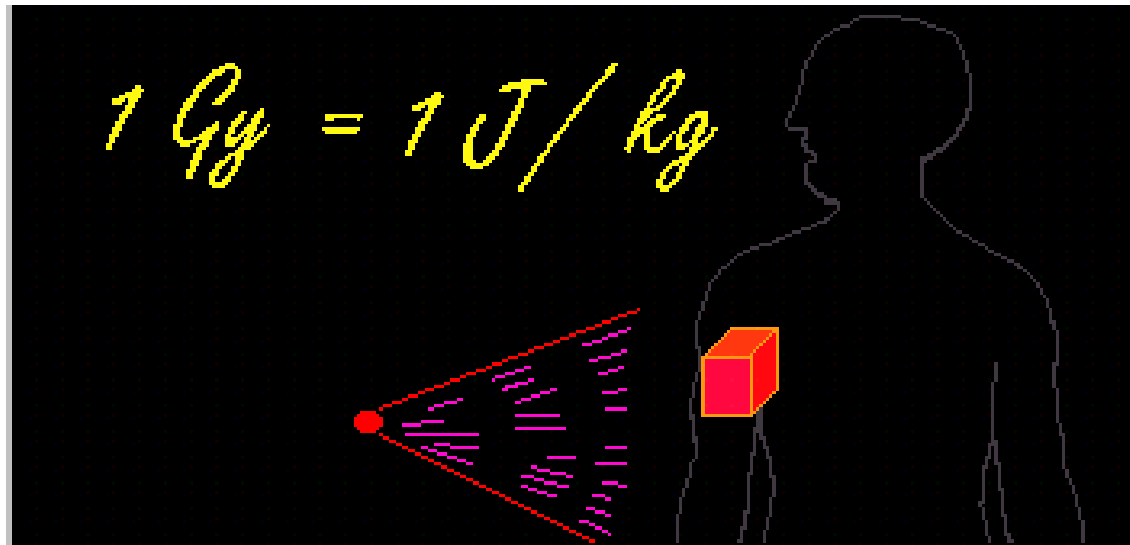
- Measurement of quantity of radiation
 - Radiation dose (Gy)
 - RBE (Relative Biological Effectiveness) / Q (Quality Factor)
 - Effective dose(Sv)

Radioactivity & Radiation Dose



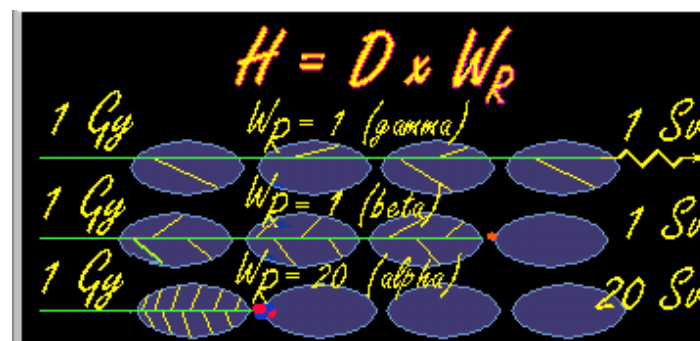
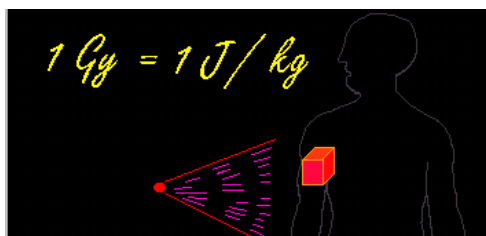
Absorbed Dose

Absorbed dose is a measure of the energy deposited in a medium by ionizing radiation



Some times it define as **KERMA** (Kinetic Energy Released in Material)

Equivalent Dose



**Equivalent
Dose (Sv)**

=

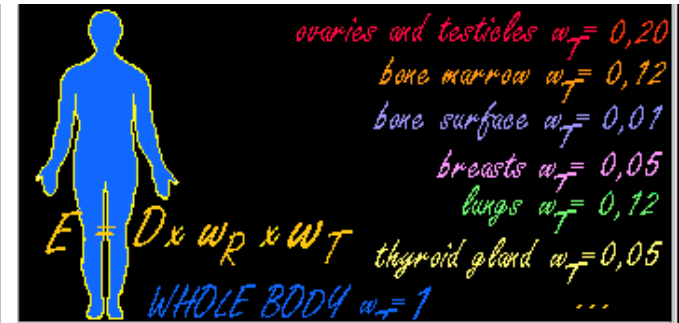
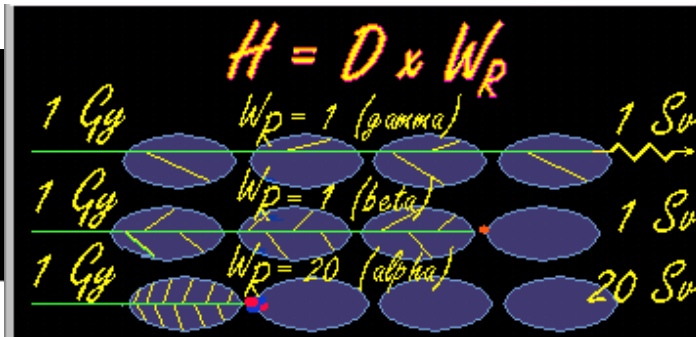
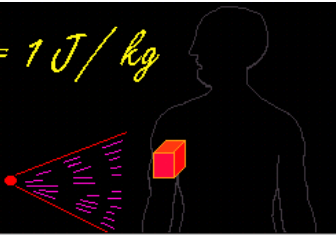
**Absorb
Dose (D)**

x

**Radiation
grating
factor (W_R)
{or Quality
actor(Q)}**

Effective Dose

$$1 \text{ Gy} = 1 \text{ J/kg}$$



$$\text{Effective Dose (Sv)} = \text{Radiation Weighting Factor (} W_R \text{)} \times \text{Tissue Weighting Factor (} W_T \text{)} \times \text{Absorbed Dose (Gy)}$$

$$\text{Effective Dose (Sv)} = \text{Quality Factor (Q)} \times \text{Relative Biological Effectiveness (RBE)} \times \text{Absorbed Dose (Gy)}$$

Type of Radiation	Radiation Weighting Factor W_R (formerly Q)
X- Ray Gamma Ray Beta Particles	1
Protones	2
Alpha Radiation	20

Organ	Tissue weighting factors W_T (formerly RBE)
<u>Gonads</u>	0.08
<u>Red Bone Marrow</u>	0.13
<u>Colon</u>	0.19
<u>Lung</u>	0.16
<u>Stomach</u>	0.12
<u>Breasts</u>	0.12
<u>Bladder</u>	0.04
<u>Liver</u>	0.04
<u>Oesophagus</u>	0.04
<u>Thyroid</u>	0.04
<u>Skin</u>	0.01
<u>Bone</u> surface	0.01
<u>Salivary glands</u>	0.01
<u>Brain</u>	0.01
Remainder of body	0.12
Total	1.00

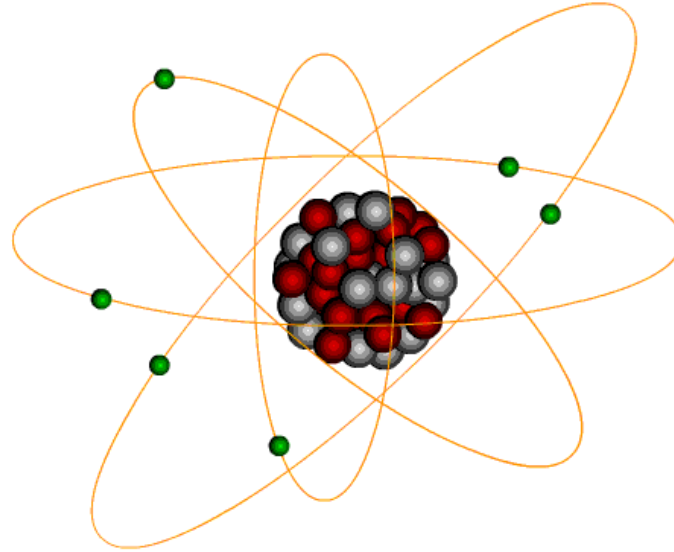
Units of Effective Dose

$$1 \text{ Sv} = 1000 \text{ mSv}$$

$$1 \text{ mSv} = 1000 \text{ uSv}$$

$$1 \text{ uSv} = 1000 \text{ nSv}$$

$$\text{Radiation Dose Rate} = \text{Sv/h}$$

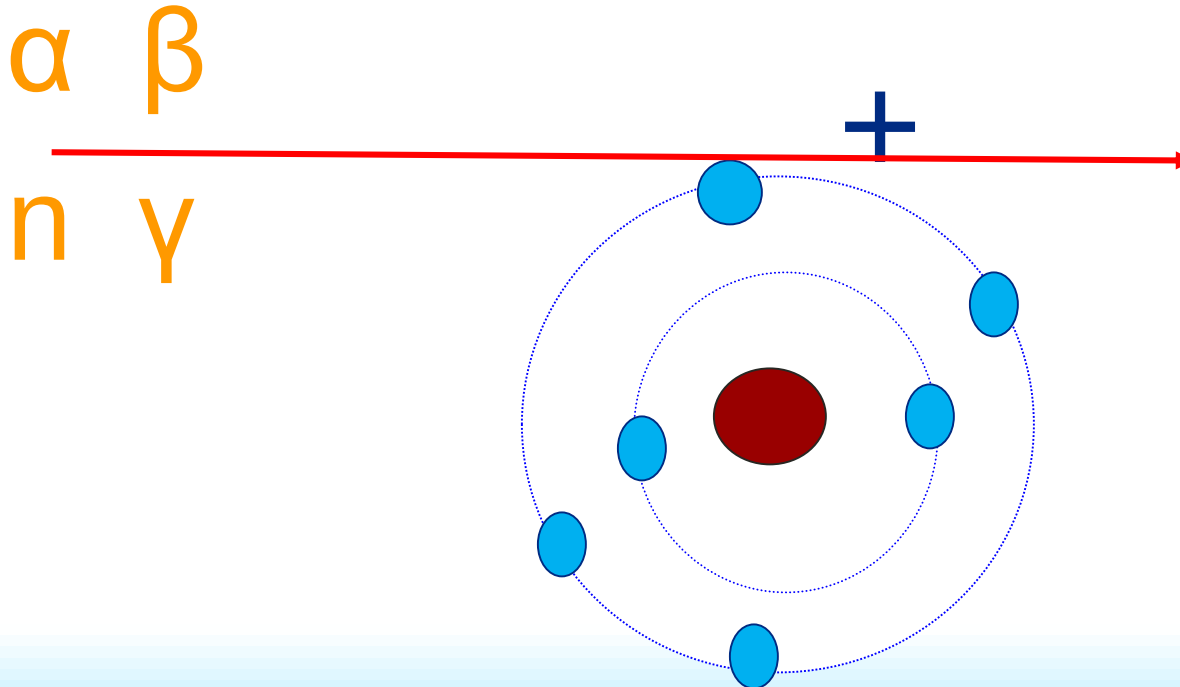


Health hazards and Radiation Protection

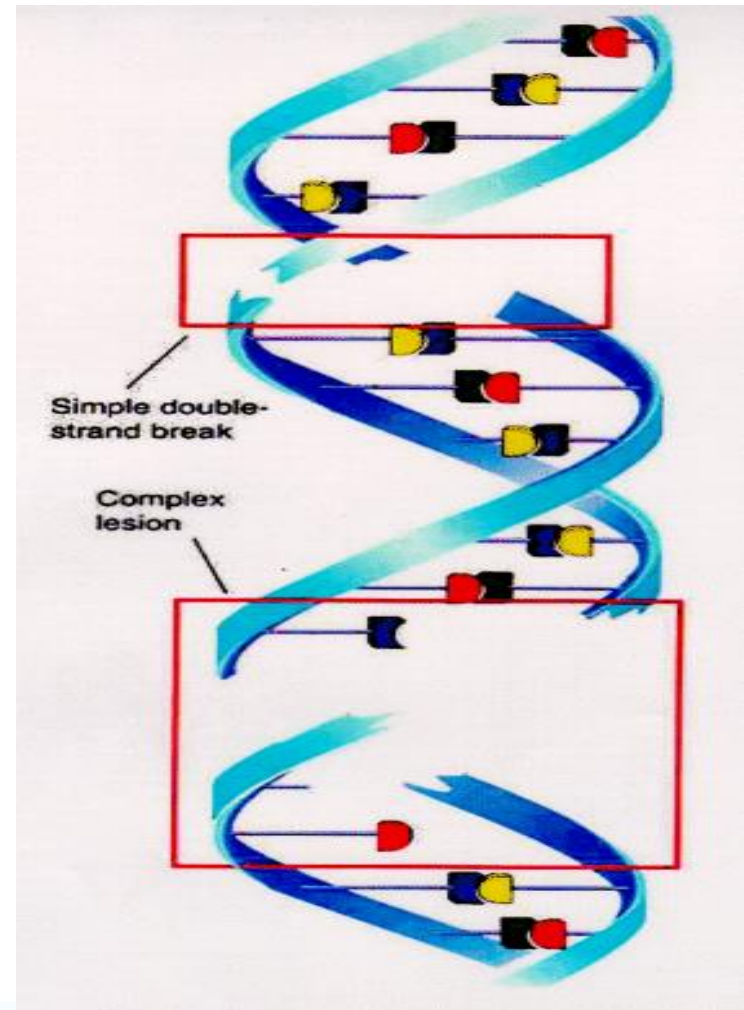
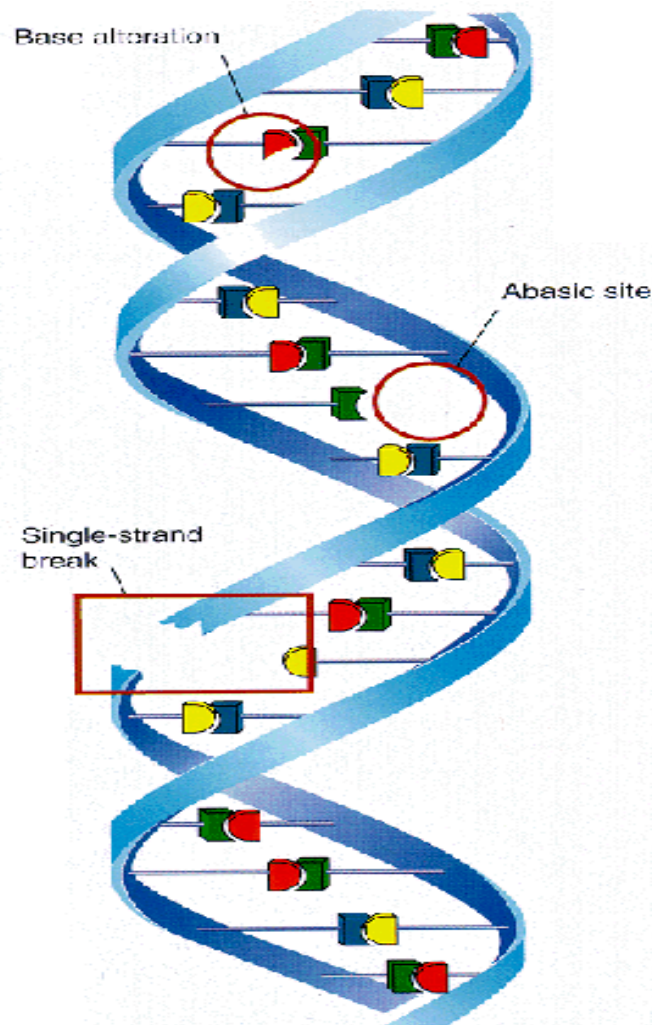
- Health hazards
 - Time of expose
 - Area of the body which exposed to radiation
 - Safety precaution

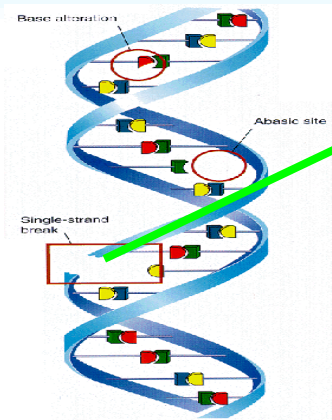
Biological Effects of Radiation

- Energy is sufficient to ionize matter
- This can lead to biological consequences
- Primary target for cell damage from ionizing radiation is DNA in the chromosomes of a cell's nucleus

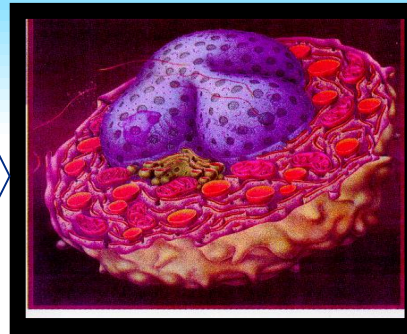


Damage of DNA Chain



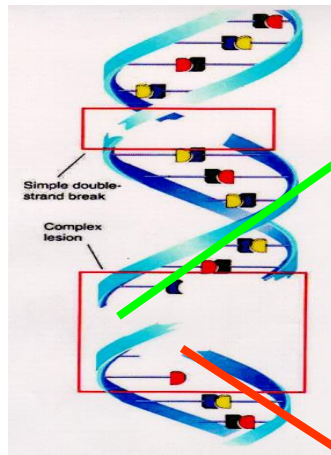


**DNA chain
repair by it
self**

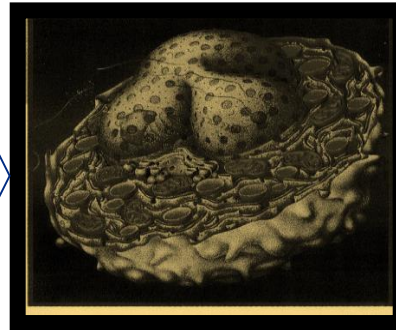


Cell survives

Viable Cell



**DNA chain
totally
death**



Cell death

Unviable Cell

High doses kill many cells

- Skin burns,
- radiation sickness
- death

Cell survives but mutated

**DNA chain
repair
incorrectly**



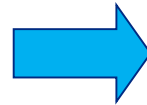
Cancer ?

Cancer induction
Hereditary
consequences

Types of Exposure & Hazards

Acute Exposure

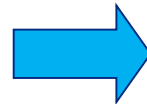
Expose to High Dose
rate within a Short time



Deterministic Effects

Chronic Exposure

Expose to small dose
rates to long times



Stochastic Effects

Deterministic Effects (Acute Exposure)

- Can only happen if the dose is above the threshold of ~
1 Sv!
- Effects are noticed shortly after exposure:
 - Burns on skin, like heavy sunburn
 - Radiation sickness
 - Blood changes: Free radicals attacking own body cells and destroying them
- The higher the dose the stronger the effect!
 - If too many cells are killed, the organ or body will die

Biological Effects: Acute and Late

Deterministic Effects (Acute)

- 1-2 Sv: vomiting, nausea, after 24 hours, no permanent damage
- 3-4 Sv: 50% deaths within 60 days
- > 8 Sv: 100% deaths in 2 weeks



Exposure & Hazards

RADIATION EFFECTS

Measurements in millisieverts (mSv). Exposure is cumulative.

- **Potentially fatal radiation sickness.**
Much higher risk of cancer later in life.

10,000 mSv: Fatal within days.

5,000 mSv: Would kill half of those exposed within one month.

2,000 mSv: Acute radiation sickness.

- **No immediate symptoms. Increased risk of serious illness later in life.**

1,000 mSv: 5% higher chance of cancer.

400 mSv: Highest hourly radiation recorded at Fukushima.
Four hour exposure would cause radiation sickness.

100 mSv: Level at which higher risk of cancer is first noticeable

- **No symptoms. No detectable increased risk of cancer.**

20 mSv: Yearly limit for nuclear workers.

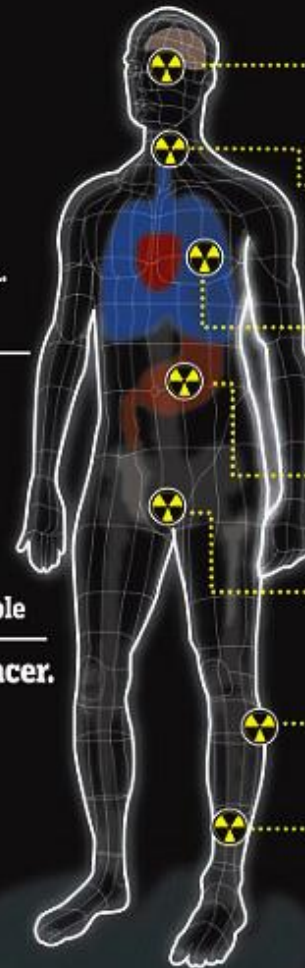
10 mSv: Average dose from a full body CT scan

9 mSv: Yearly dose for airline crews.

3 mSv: Single mammogram

2 mSv: Average yearly background radiation dose in UK

0.1 mSv: Single chest x-ray



EYES High doses can trigger cataracts months later.

THYROID Hormone glands vulnerable to cancer. Radioactive iodine builds up in thyroid. Children most at risk.

LUNGS Vulnerable to DNA damage when radioactive material is breathed in.

STOMACH Vulnerable if radioactive material is swallowed.

REPRODUCTIVE ORGANS
High doses can cause sterility.

SKIN High doses cause redness and burning.

BONE MARROW Produces red and white blood cells. Radiation can lead to leukaemia and other immune system diseases.

Exposure vs. Contamination

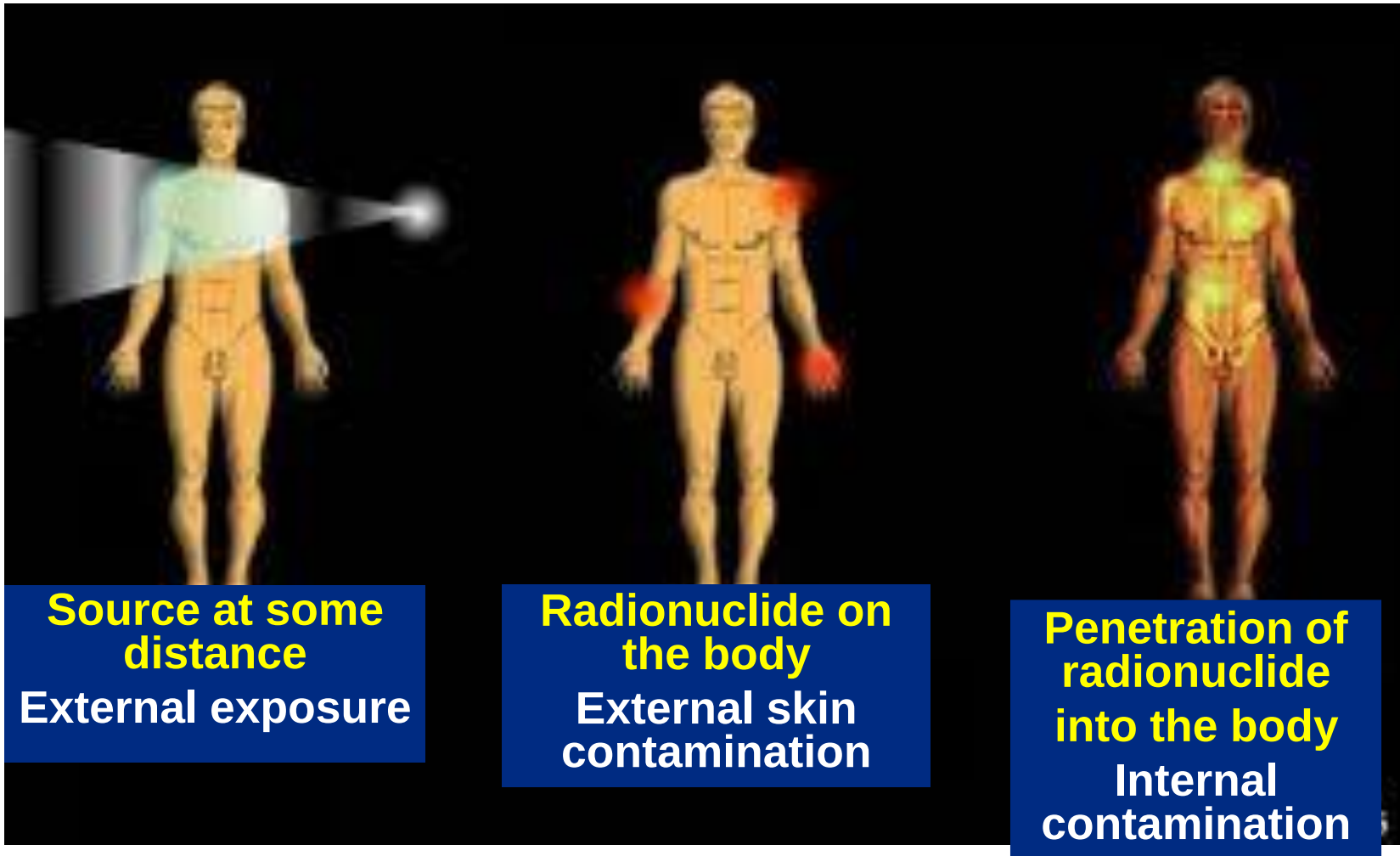
Exposure

- If you are near a source, you are exposed
- The longer you stay, the closer you get, the higher your radiation dose
- You do not become radioactive

Contamination

- If radioactive material gets on you, or inside you, you are contaminated
- You are radioactive and will be continuously irradiated at close distances or internally, until the contamination has been removed

Exposure vs. Contamination



Natural Radioactivity

Cosmic Radiation

Cosmogenic Radionuclides

Ex: ^3H , ^7Be , ^{14}C

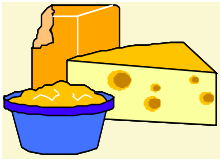
Primordial Radionuclides

Ex: ^{238}U , ^{232}Th ----- ^{235}U , ^{226}Ra , ^{222}Rn , ^{230}Th , ^{228}Th
 ^{40}K

Annual Radiation Exposure



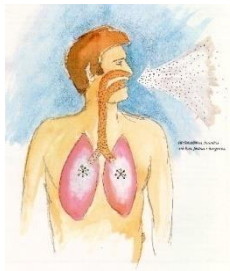
By Cosmic Radiation: 0.4 mSv/year



By Food: 0.3 mSv/year



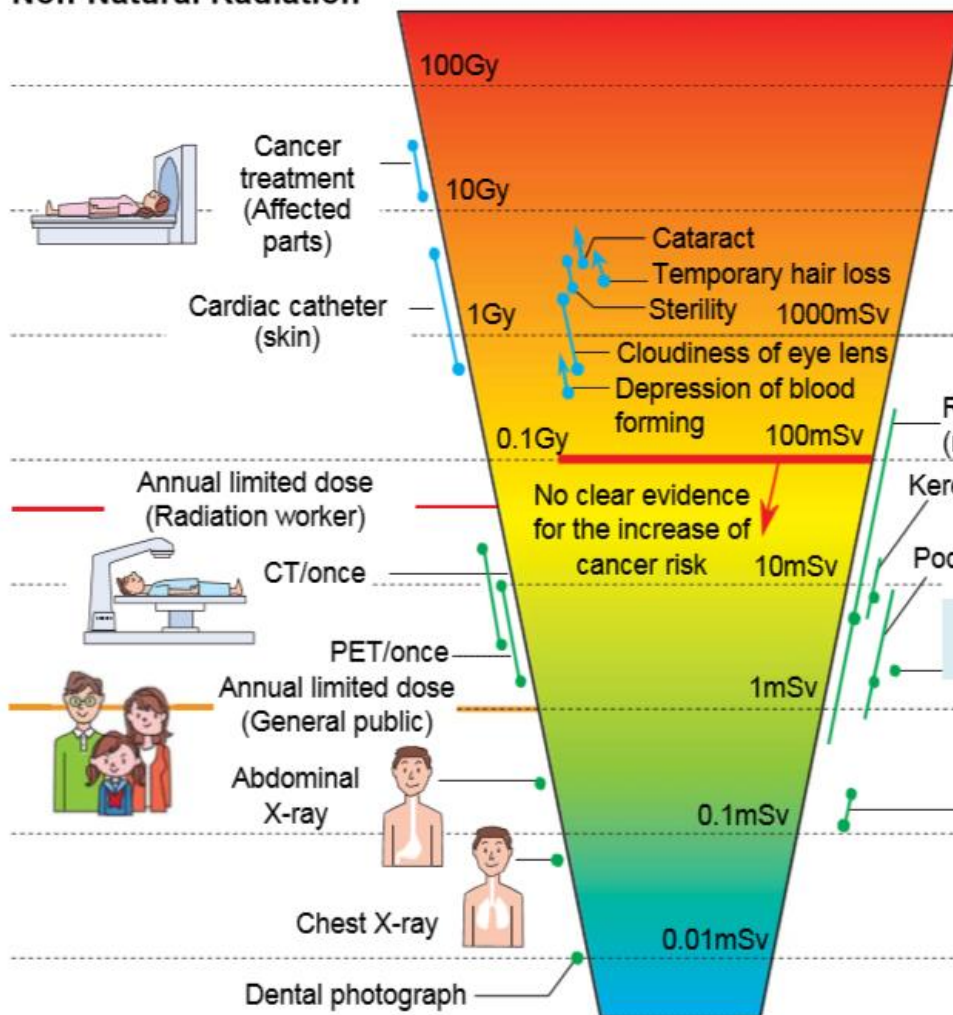
By Surrounding Environment: 0.4 mSv/year



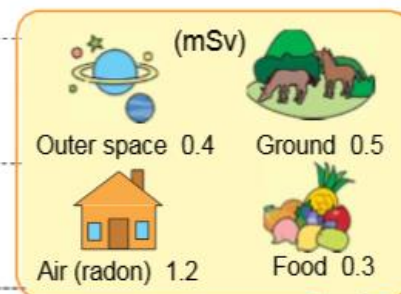
By Inhalation: 1.3 mSv/year

Total : 2.4 mSv/year

Non-Natural Radiation



Natural Radiation



Ramsar, Iran
(natural radiation/year)

Kerela and Chennai, India

Pocos de Caldas, Brazil

World: Natural Radiation
(2.4 mSv/person/year)

Tokyo-NY(Round trip)
(the increase of cosmic rays due to altitude)

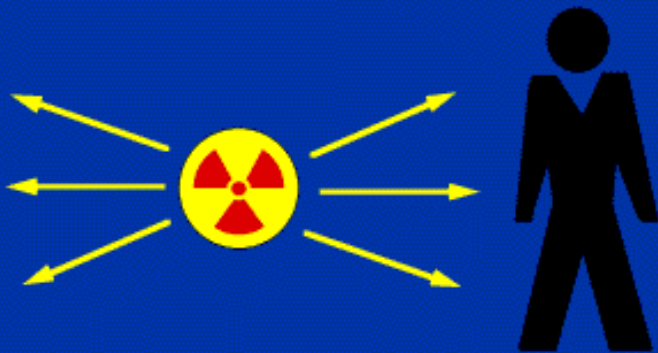
Units of radiation dose

Gray (Gy) : The amount of radiation energy absorbed by materials and humans

Millisievert (mSv) : Radiation risks (cancer, hereditary effects*) on the human body

Two Exposure Ways

External Exposure



Internal Exposure

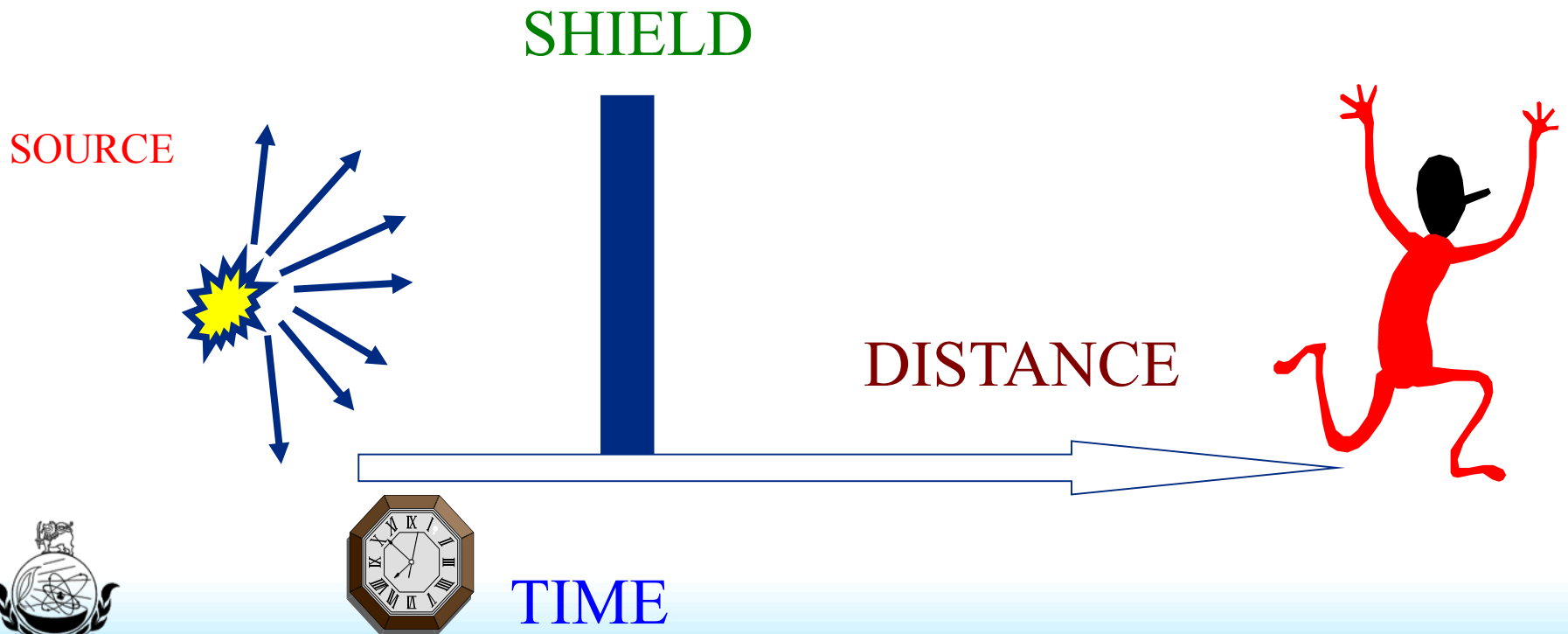


Exposure from External
Radiation Source (Gamma or
Beta)

Exposure from Internal
Radiation Source (Alpha, Beta
or Gamma)

Principles of Control External Exposure

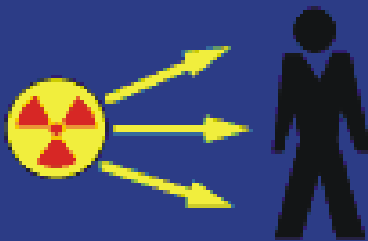
- Reduce the Exposure to your body
 - **Time:** reduce time next to source of radiation
 - **Distance:** increase distance from source of radiation
 - **Shielding:** use appropriate shielding



Reduce Time

Time Relationship

Exposure rate
= 1 mR/hr $\times$ Time = Total Exposure

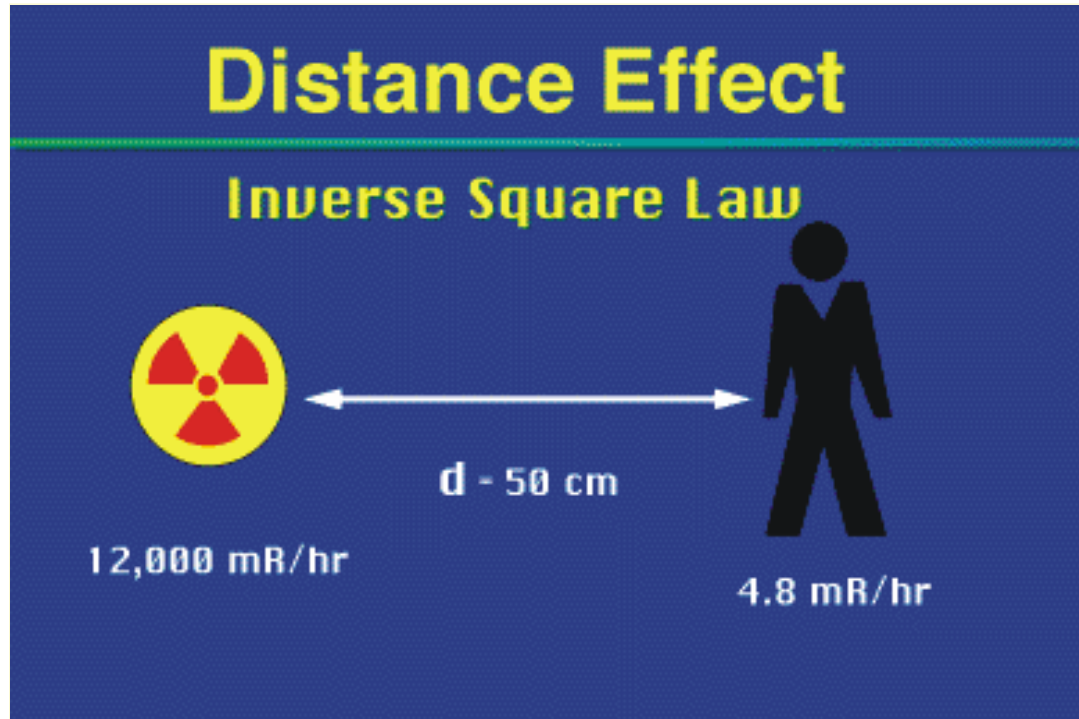


1 hour = 1 mR



2 hours = 2 mR

Increase Distance)



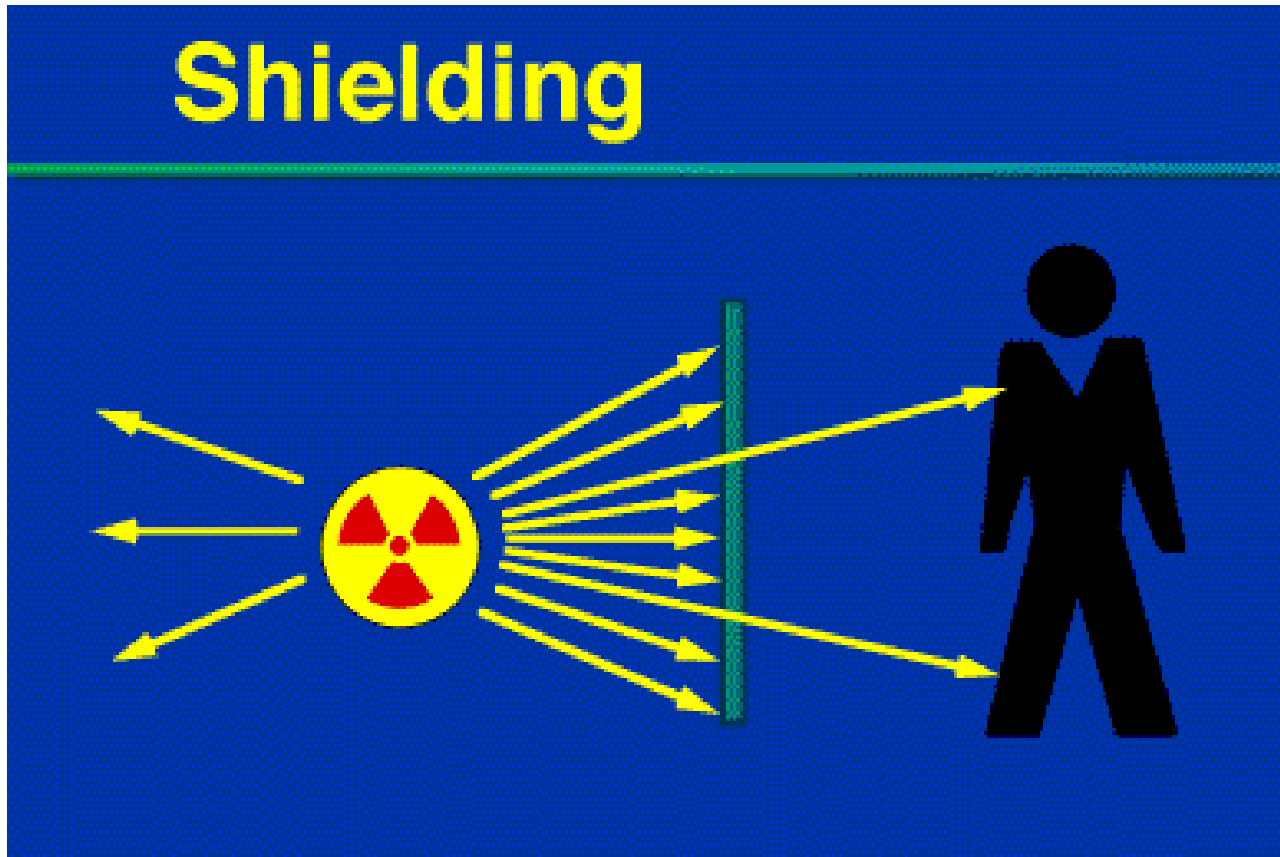
For a Point Source

$$R = \frac{k}{d^2}$$

R = Radiation Dose Rate
d = Distance to Radiation
Source

K = Constant

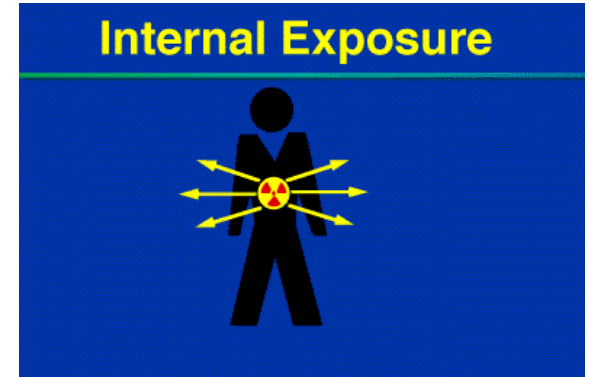
Shielding

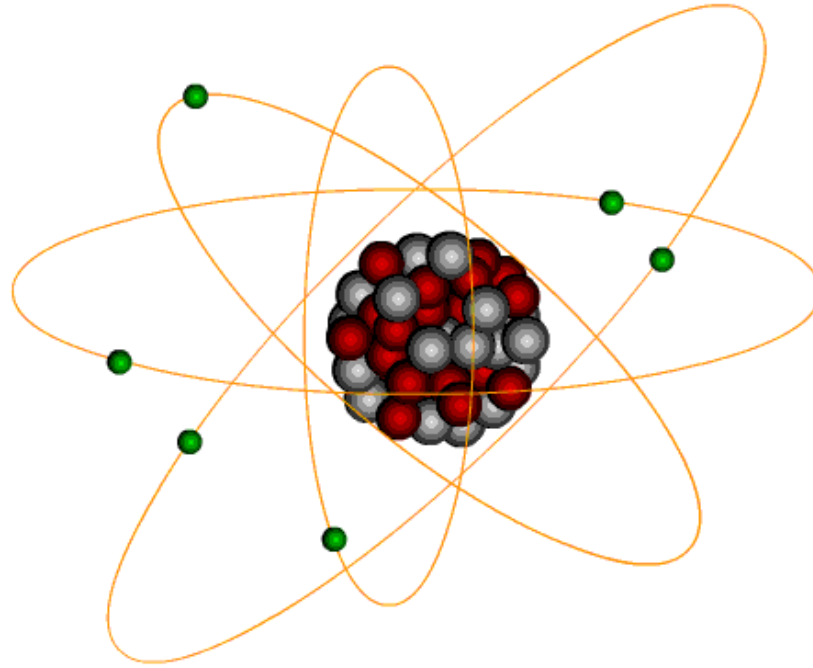


Shielding Type depends on the Radiation Type

Principles of Control Internal Exposure

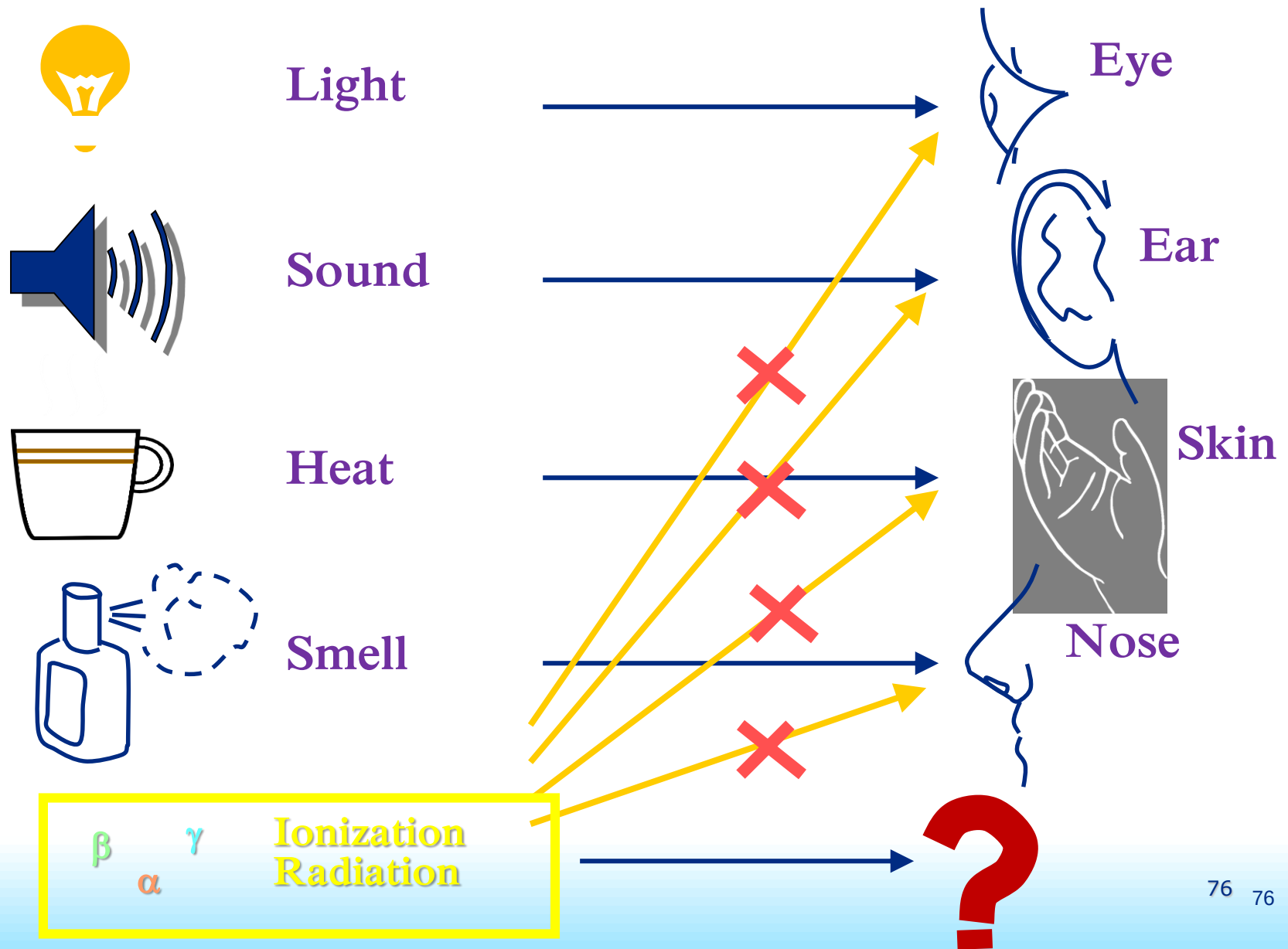
- **Avoid from Contamination**
 - Never touch a radioactive material with your hands !!
 - Always use Protective cloths when work with unshielded sources (Mask, Gloves, Overall etc..)
- **If contamination is suspected:**
 - Change your clothes immediately and have them thoroughly washed.
 - Wash your hands and body with warm soapy water.





Radiation Measurements

Ionization Radiation & Sensitivity



How we detect?

- Ionization Radiation is not sensitive to any organ
- Radiation Measuring Instrument should be used

Survey meter



Detector

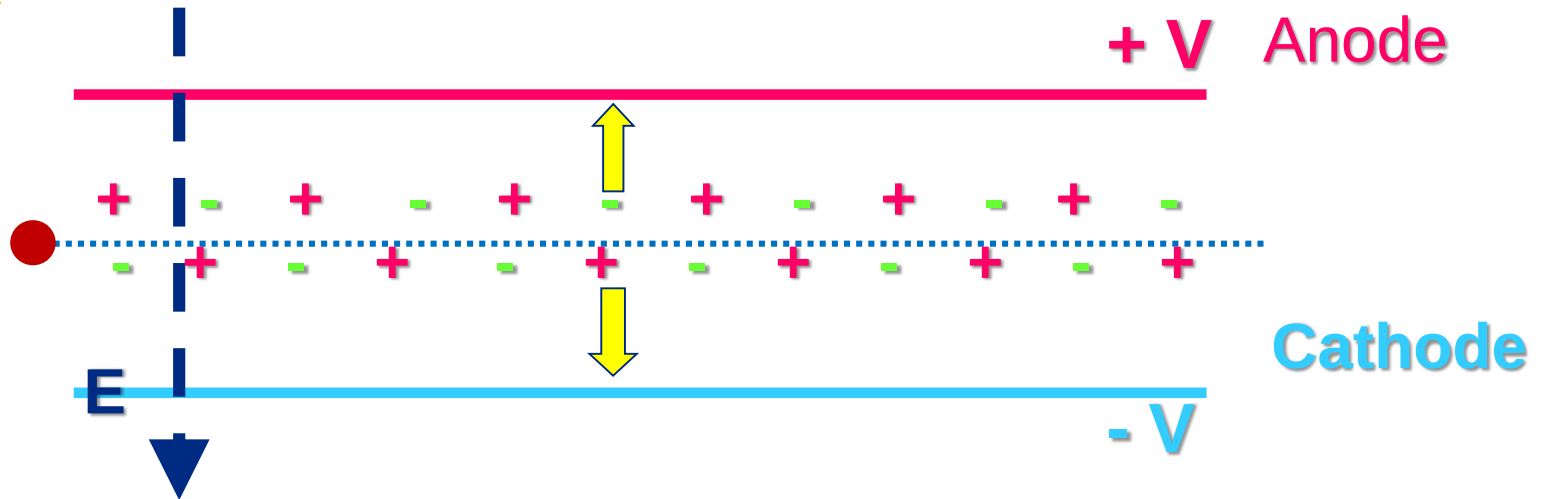
Radiation Detectors

- Developed with Various Physics Theory
 - GM counters
 - Ionization Chambers
 - Proportional Counters
 - Solid State Scintillation Detectors
 - Semiconductor Detectors



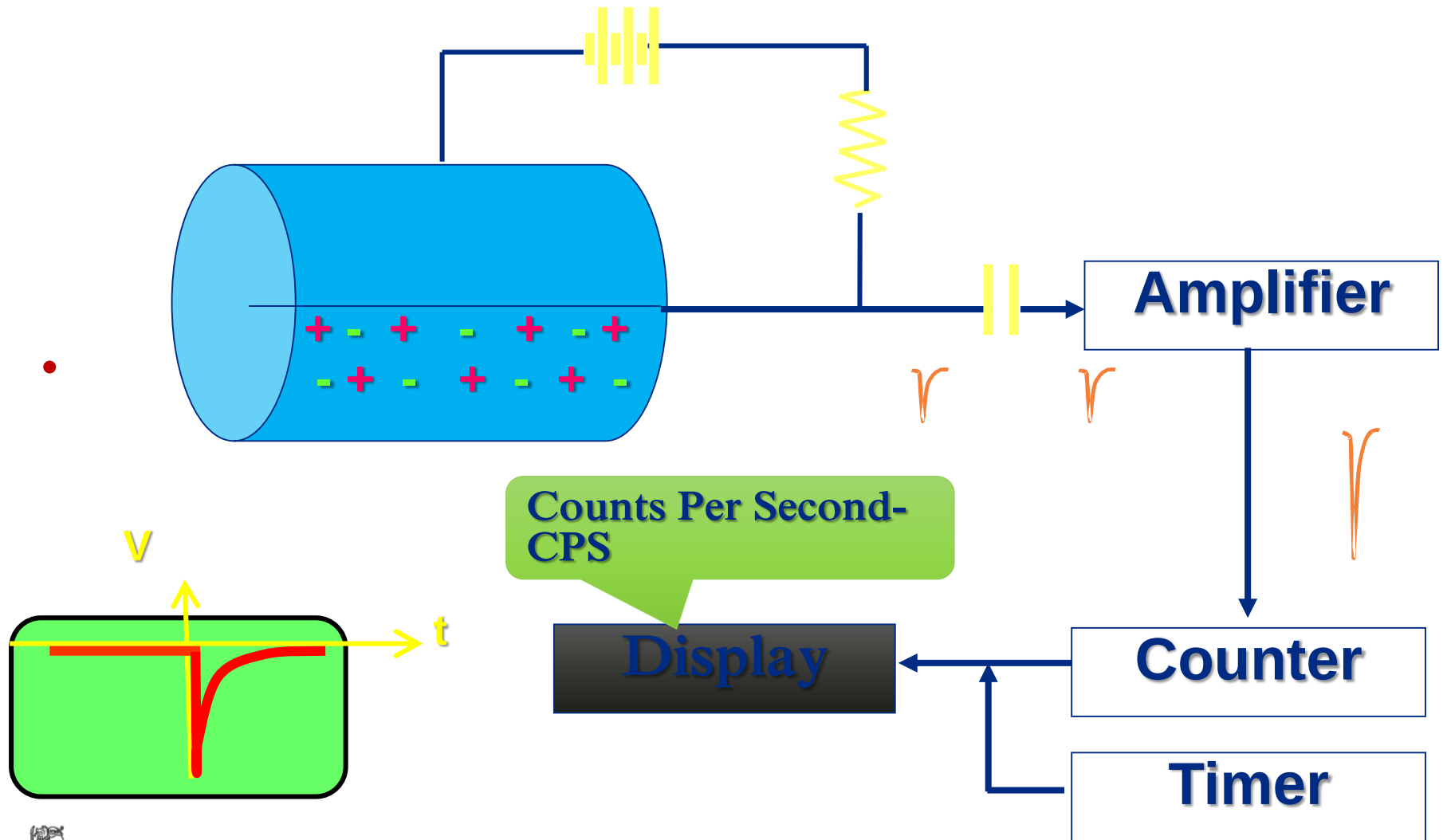
Working Principle of Radiation Detector

Principle of GM Tube



- Primary Ion Pairs Generate with Ionizing Radiation
- Acceleration due to Applied voltage
- Create Secondary Ion Pairs
- Total Charge Generate Electrical pulse

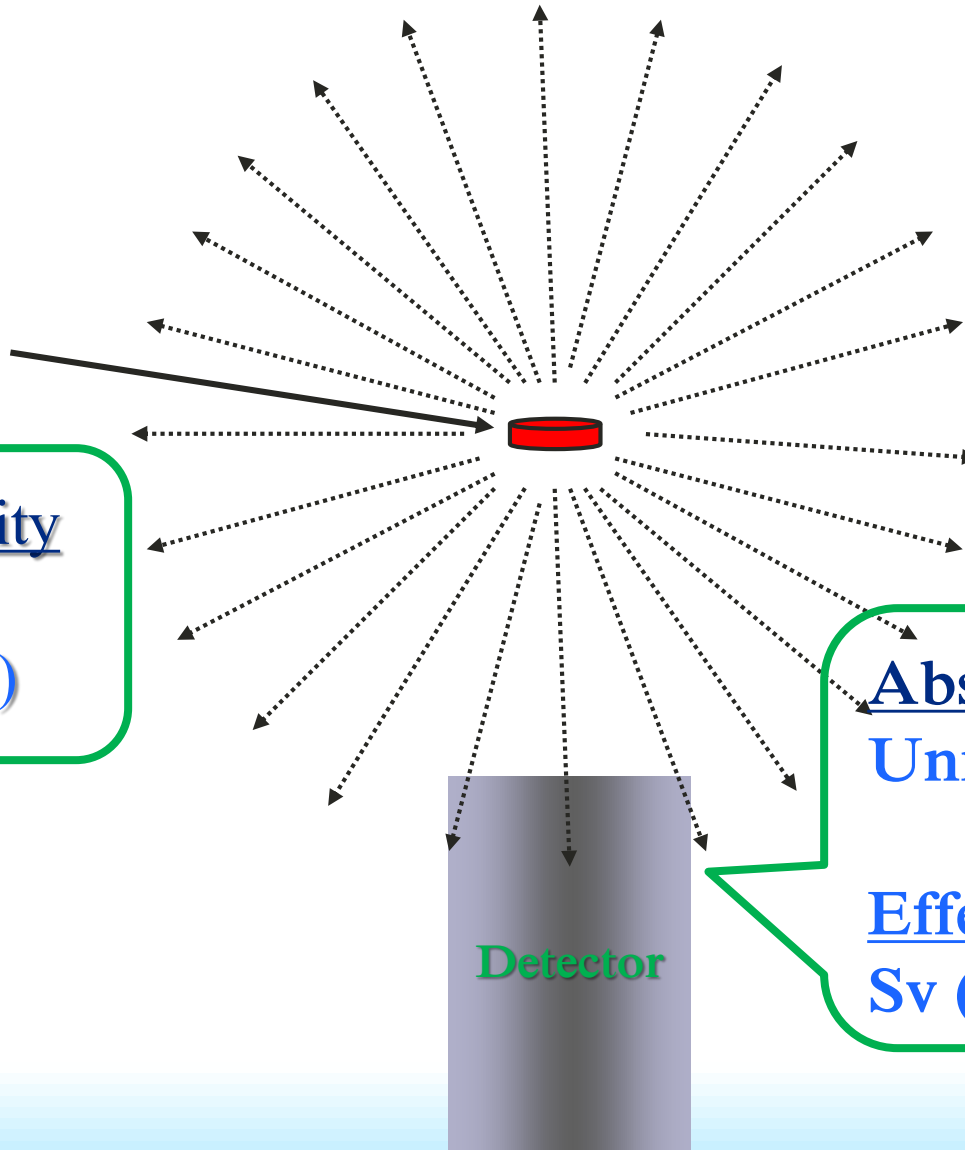
Working Principle of Survey Meter



Radiation Measuring Units

Source

Radioactivity
Unit – Bq
(Becquerel)



Absorb Dose
Unit – Gy (Gray) = J/kg

Effective Dose
Sv (Svert)

Reading of Survey meter

**Counts Per
Second
CPS**

X

Factor

=

**Radiation Dose
uSv/h**

4 CPS

- Enough to Identify Radiation Intensity strength at a Place

50 uSv /h

- Radiation Dose can be Identified

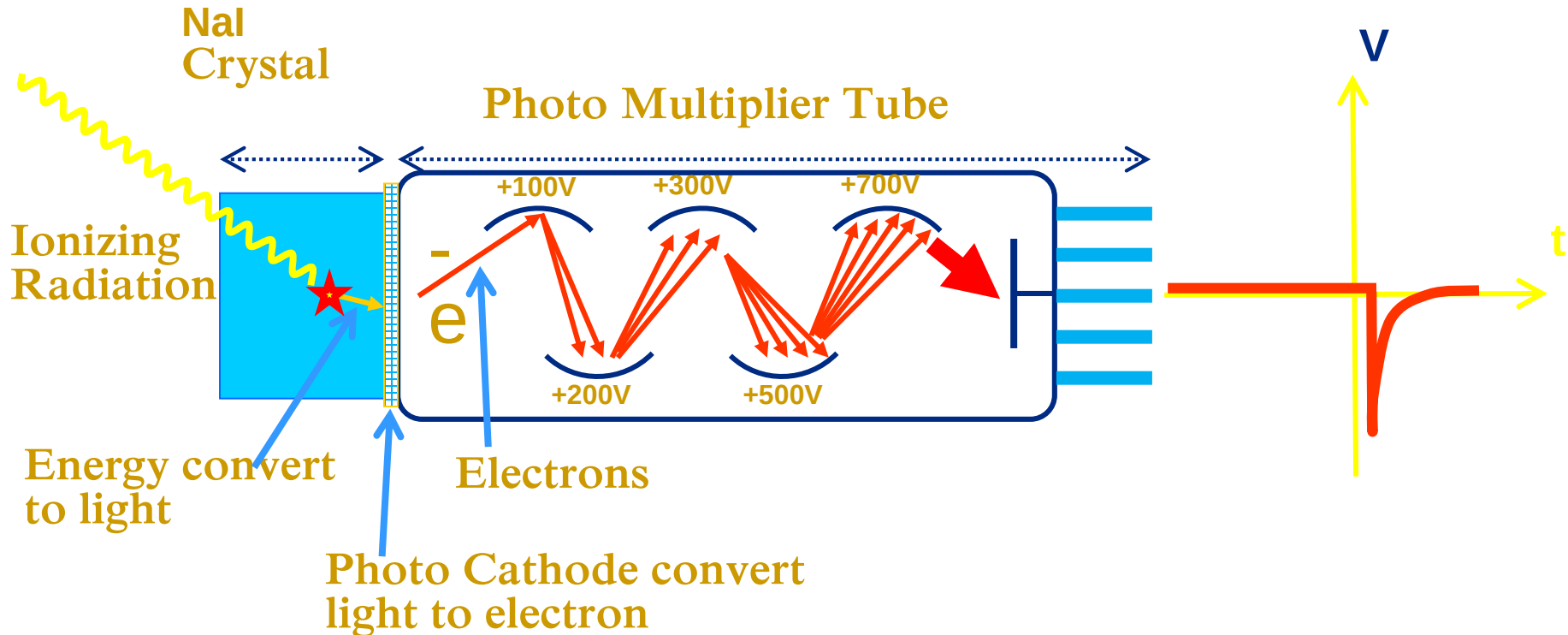
Identification of Radionuclides

- Radionuclide can be identified with its unique gamma photon energy

Radionuclide	Its gamma photon energy
Cs -137	662 keV
Co-60	1173 keV and 1332 keV
K-40	1460 keV
I -131	364 keV, 637 keV, 284 keV

- Electronic pulse Generate by the Detector should be proportional to the Gamma Energy Photon

Scintillation Detector

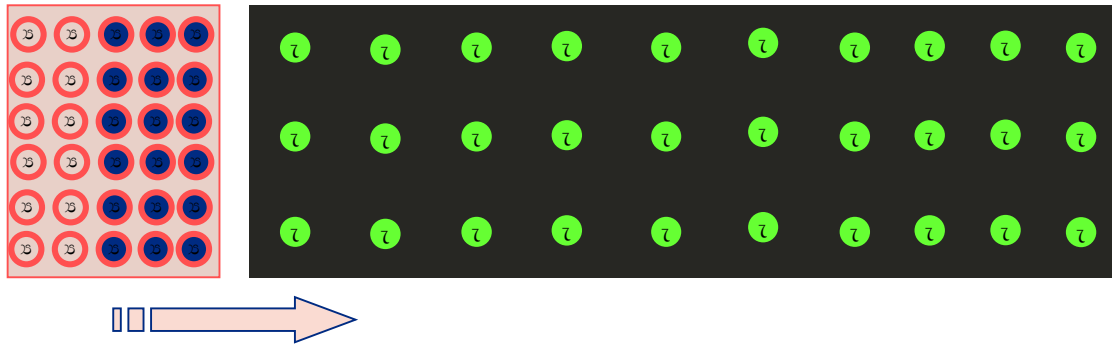


Pulse Height proportional to Gamma Ray Energy

Number of pulse proportional to Radiation Intensity

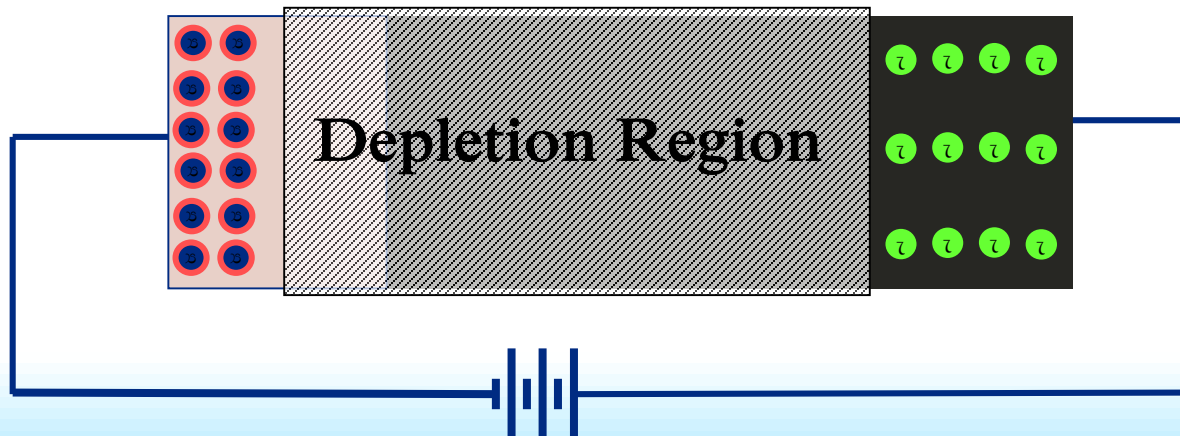
HpGe Detector

p-n Junction of a HpGe Detector



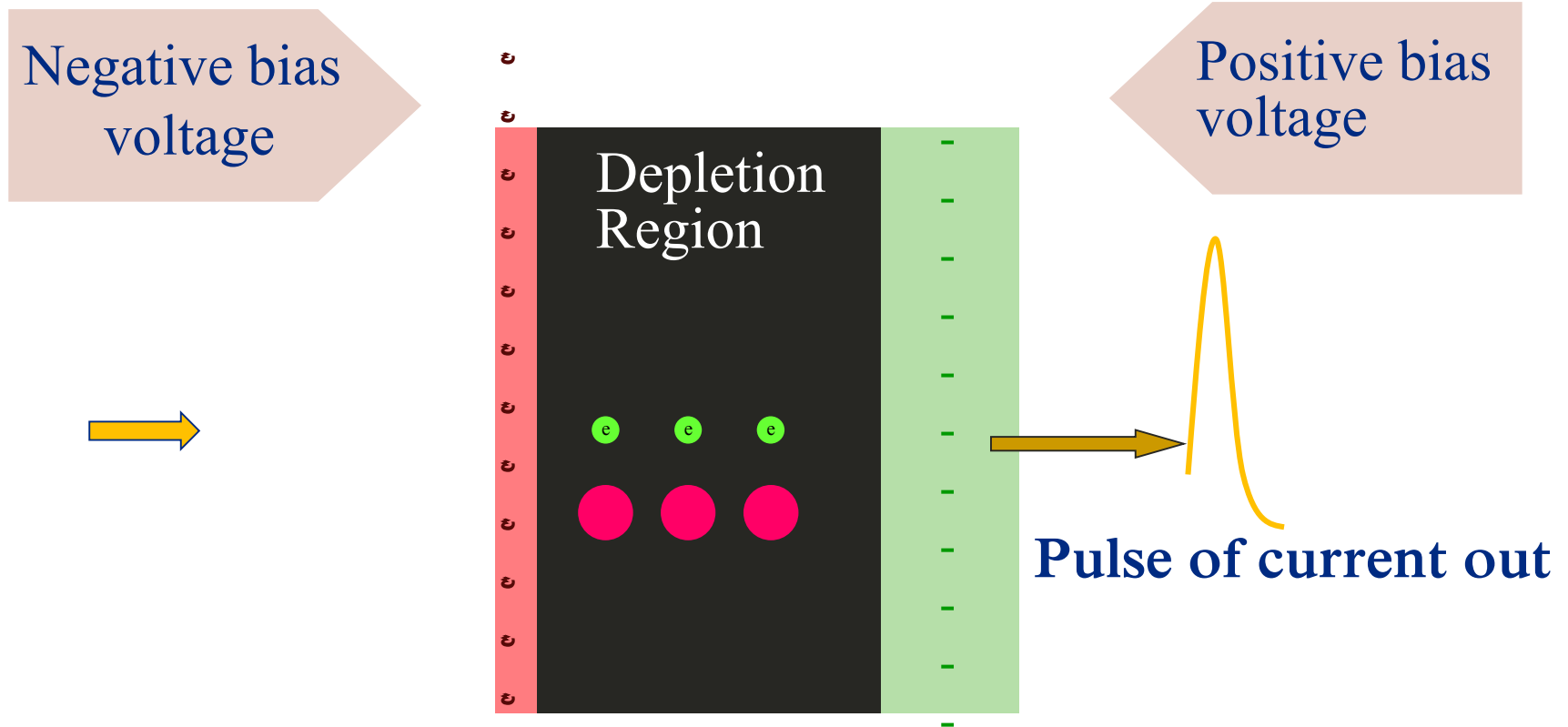
Reversed Biased Diode

Negative
bias
voltage



Positive
bias
voltage

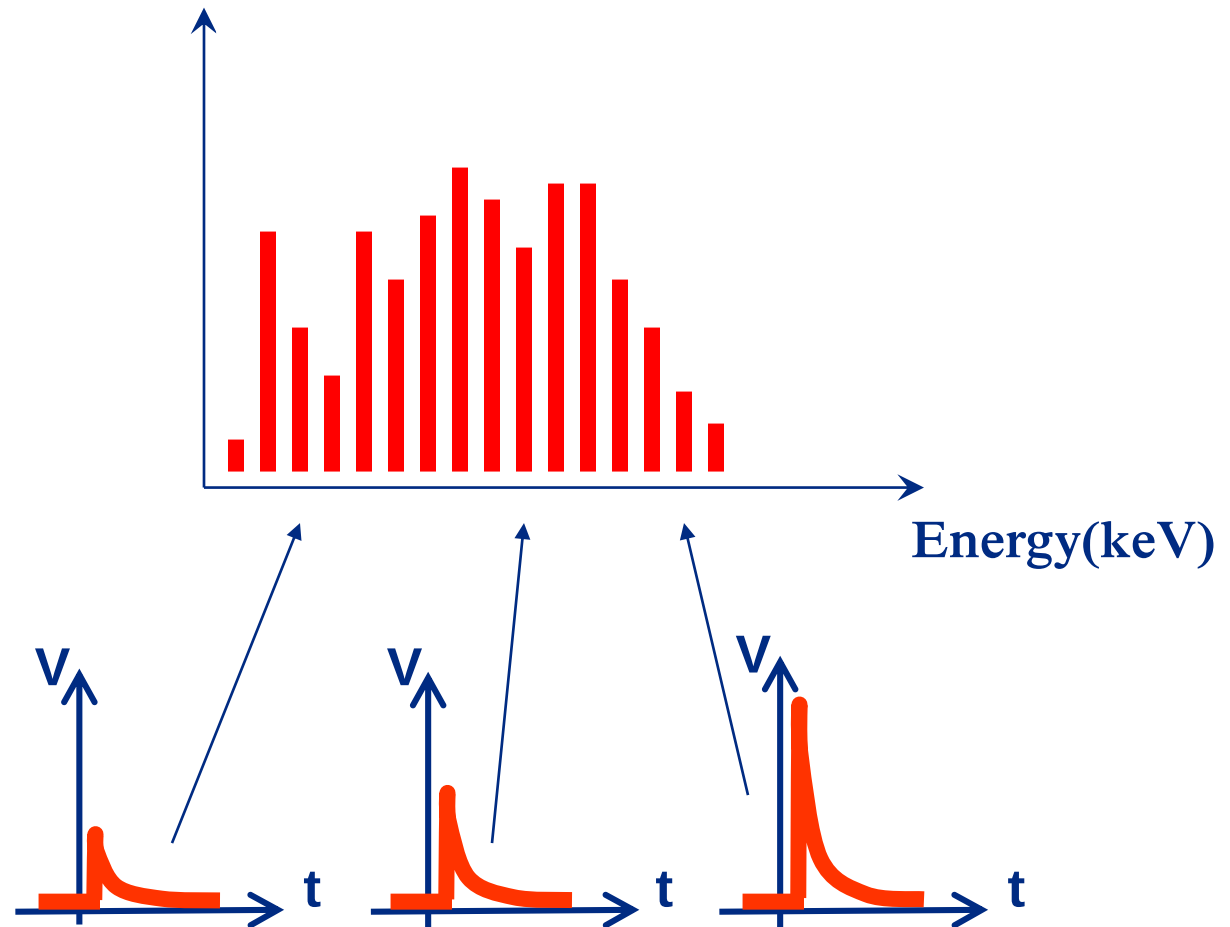
HpGe Detector



Liquid nitrogen is used to cool down detector to room temperature

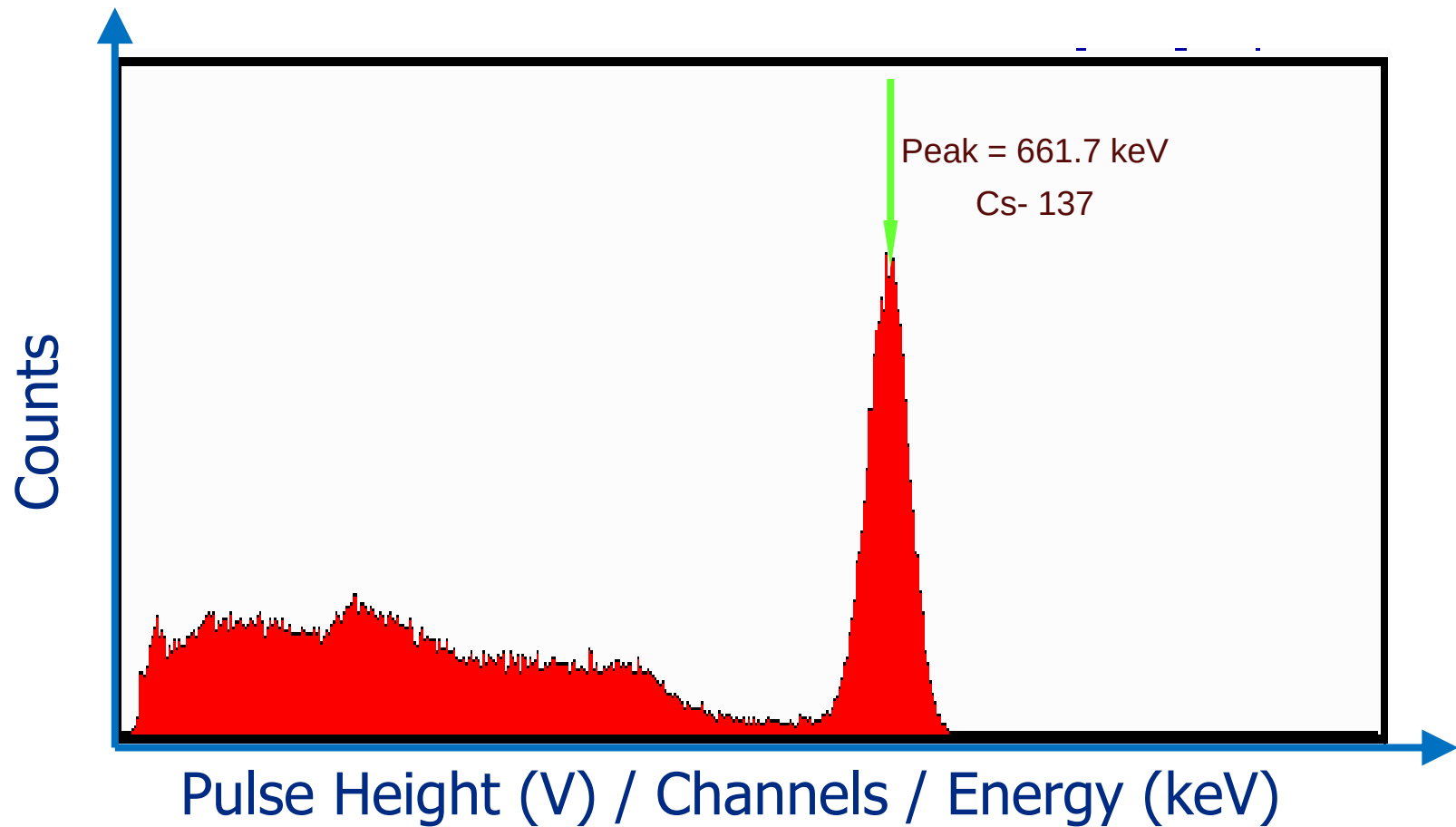
Obtain Energy Spectrum

Number of
Pulses

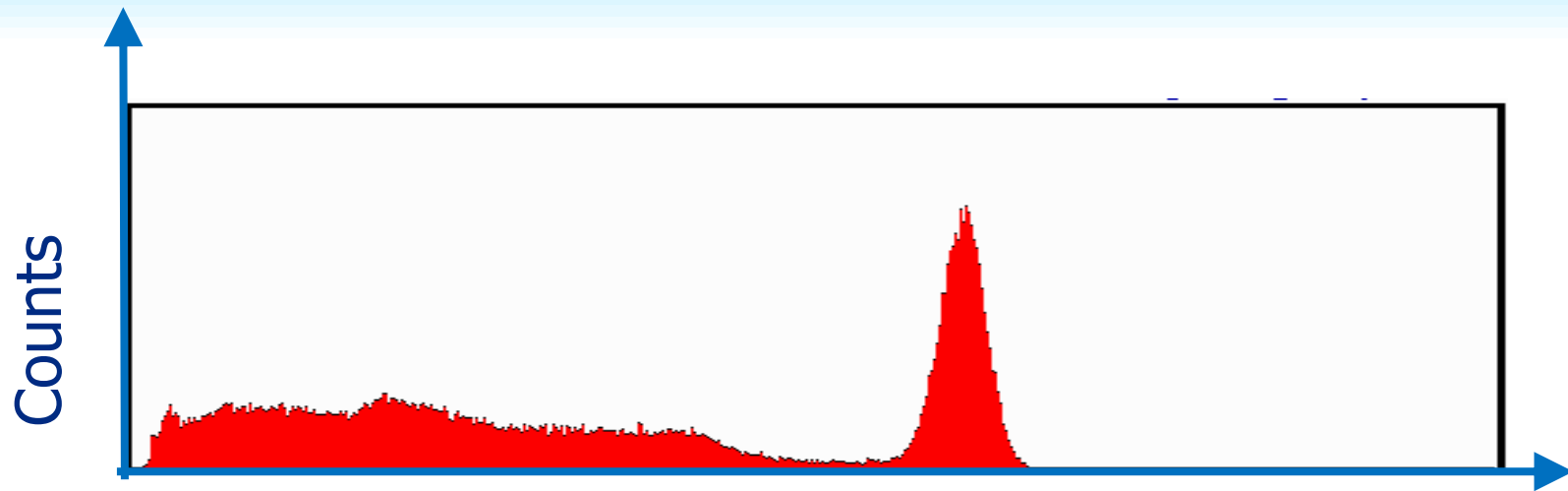


Pulse Height $\propto$ Energy

Obtain Energy Spectrum



Obtain Energy Spectrum



Pulse
Height (V)



Channels

2	6	5	9	8	7	5	2	40	38	1	0	0	0	0
---	---	---	---	---	---	---	---	----	----	---	---	---	---	---

1024

Energy

1
keV

2000 keV

Comparison between Spectrum

